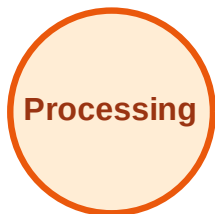


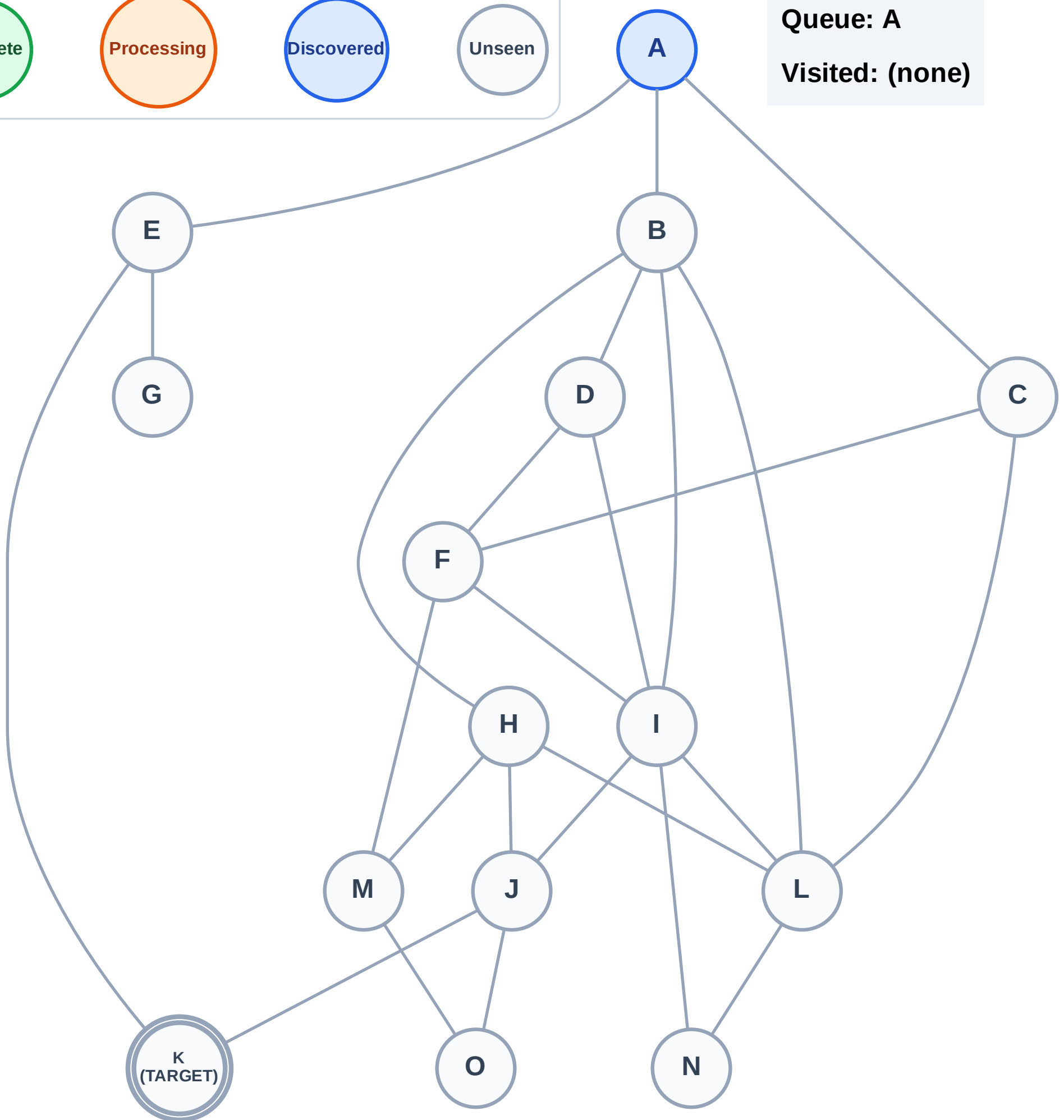
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Queue: A
Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

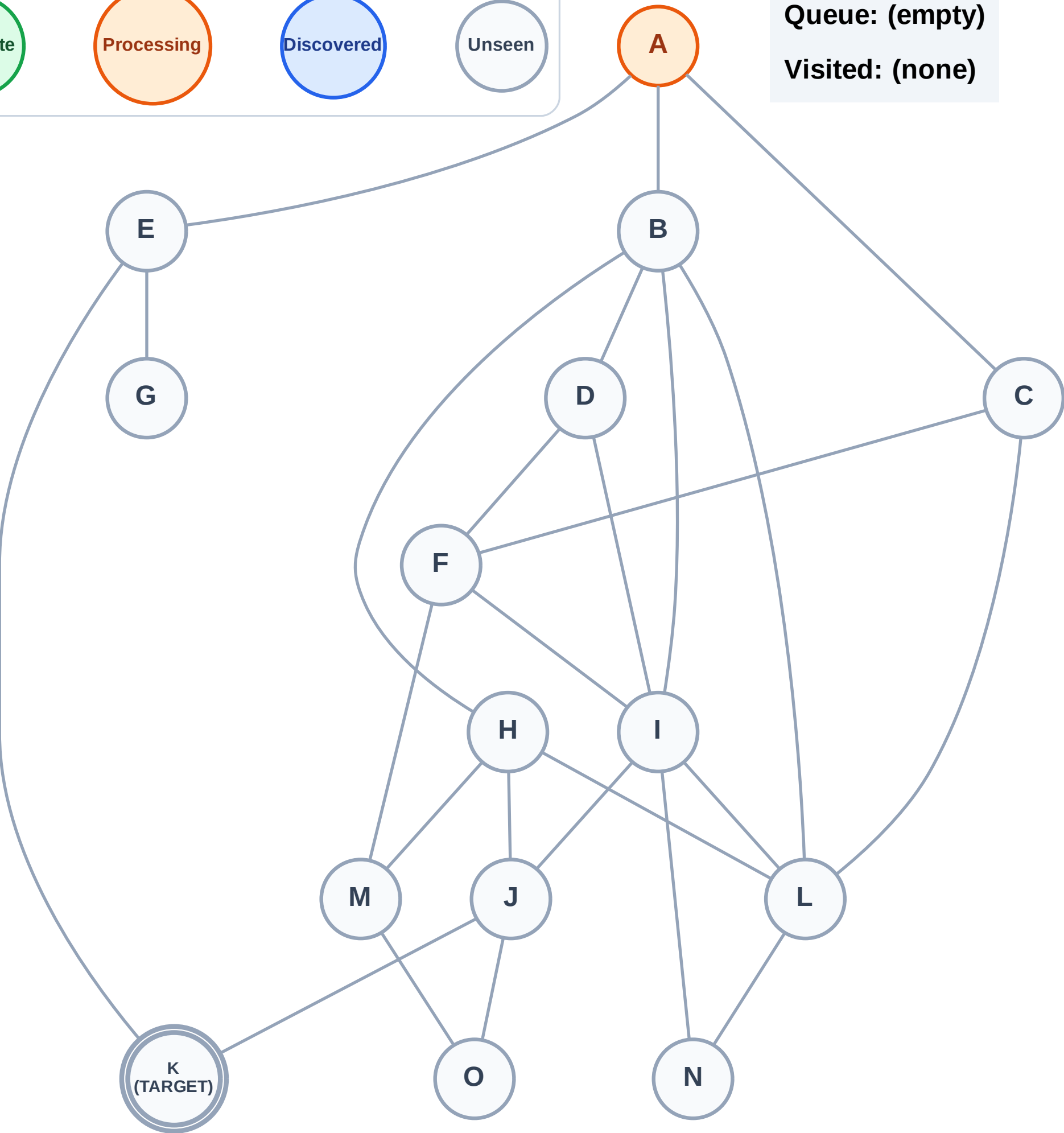
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, C, E from A

Legend

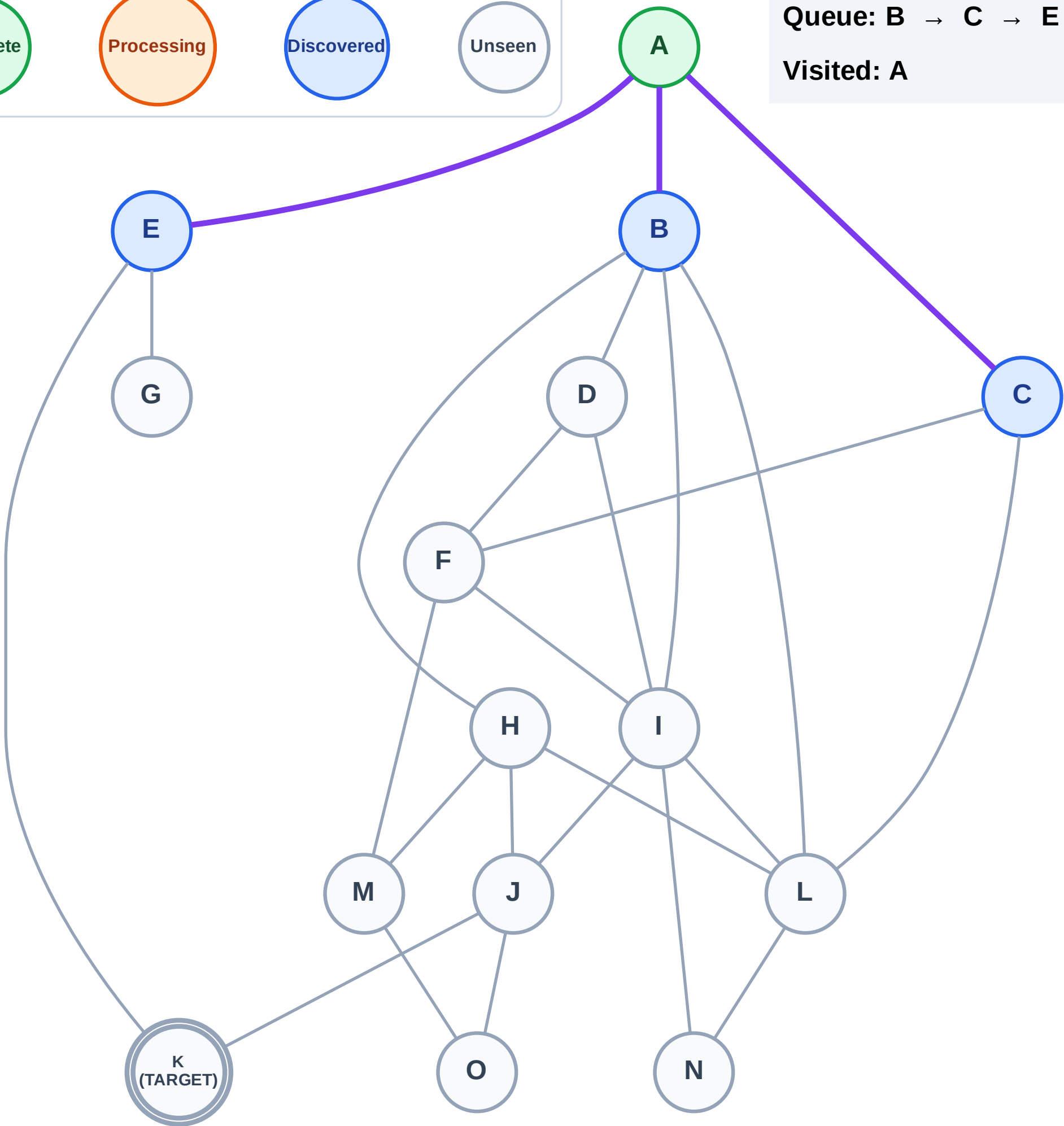
Complete

Processing

Discovered

Unseen

Queue: B → C → E
Visited: A



BFS Traversal

Step 4: Process node B

Legend

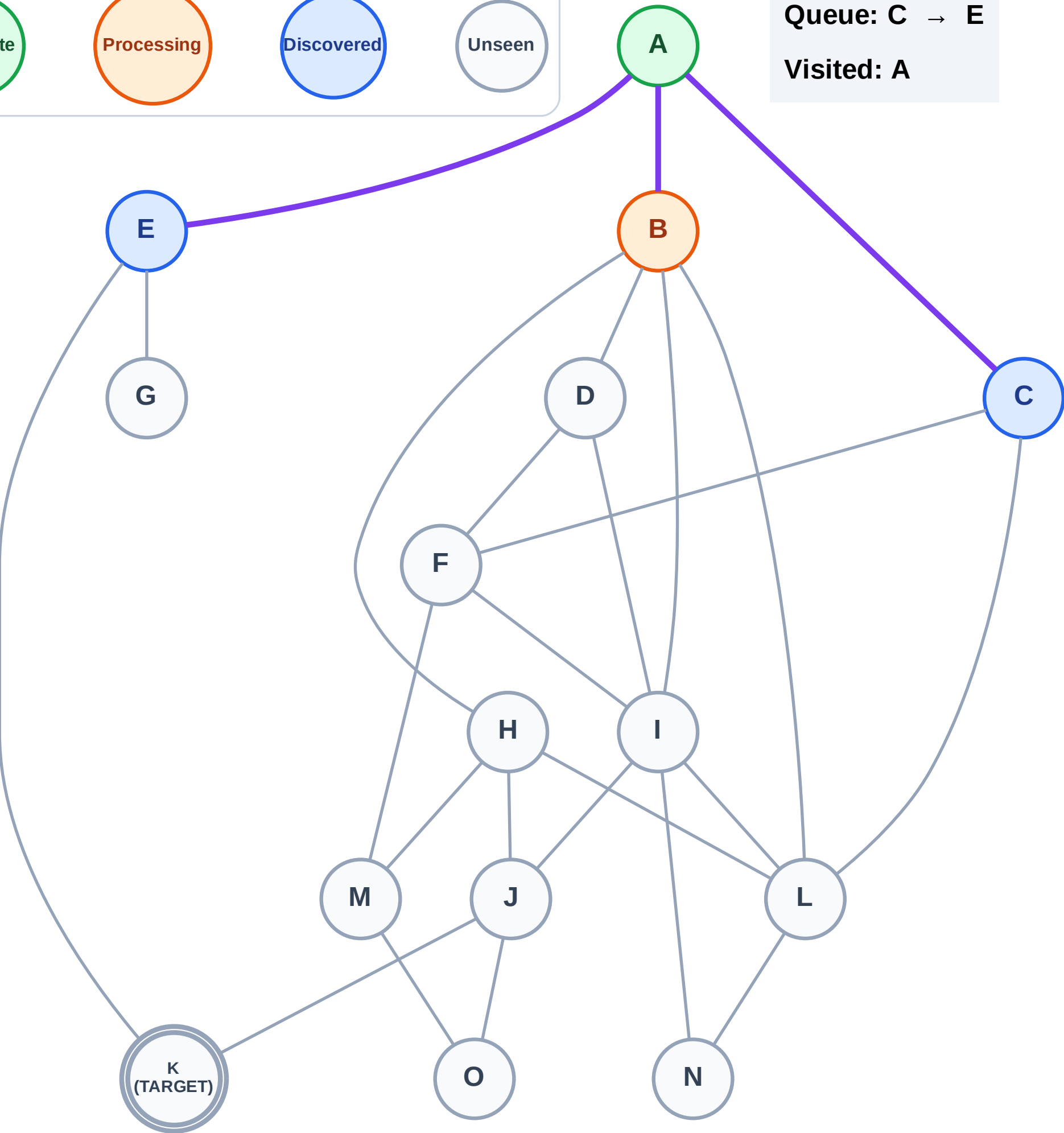
Complete

Processing

Discovered

Unseen

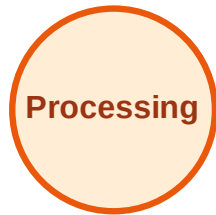
Queue: C → E
Visited: A



BFS Traversal

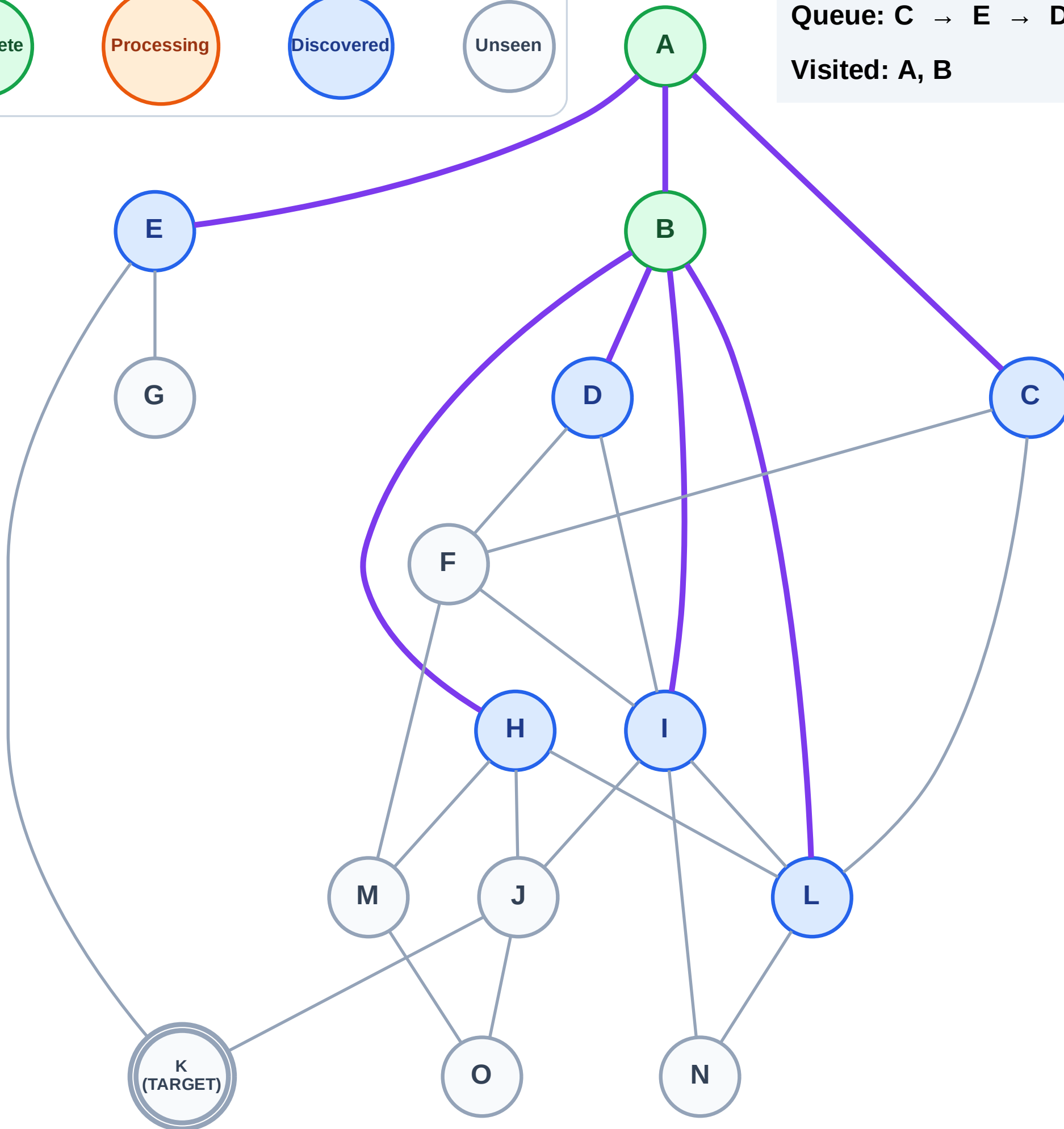
Step 5: Discover D, H, I, L from B

Legend



Queue: C → E → D → H → I → L

Visited: A, B



BFS Traversal

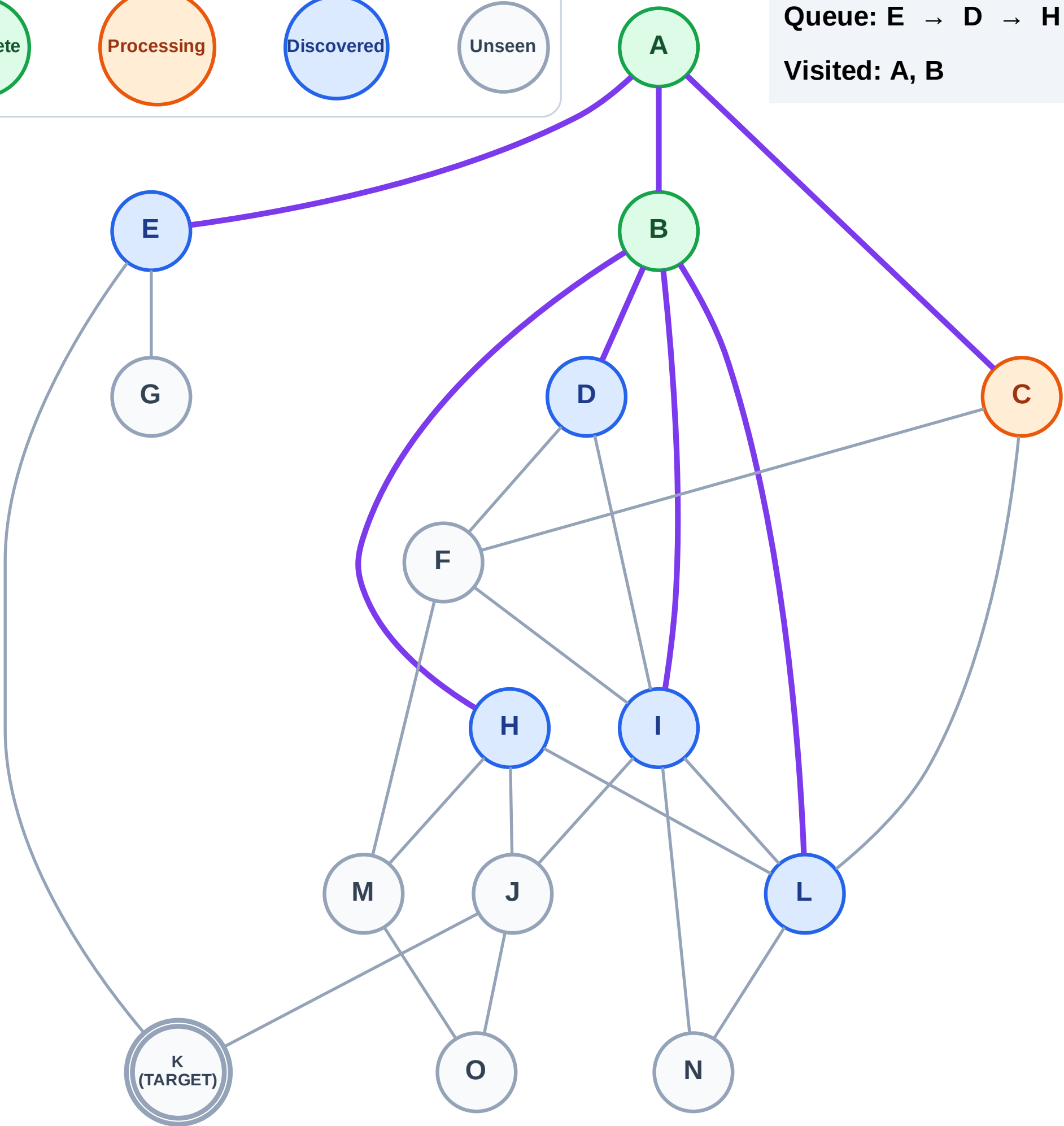
Step 6: Process node C

Legend



Queue: E → D → H → I → L

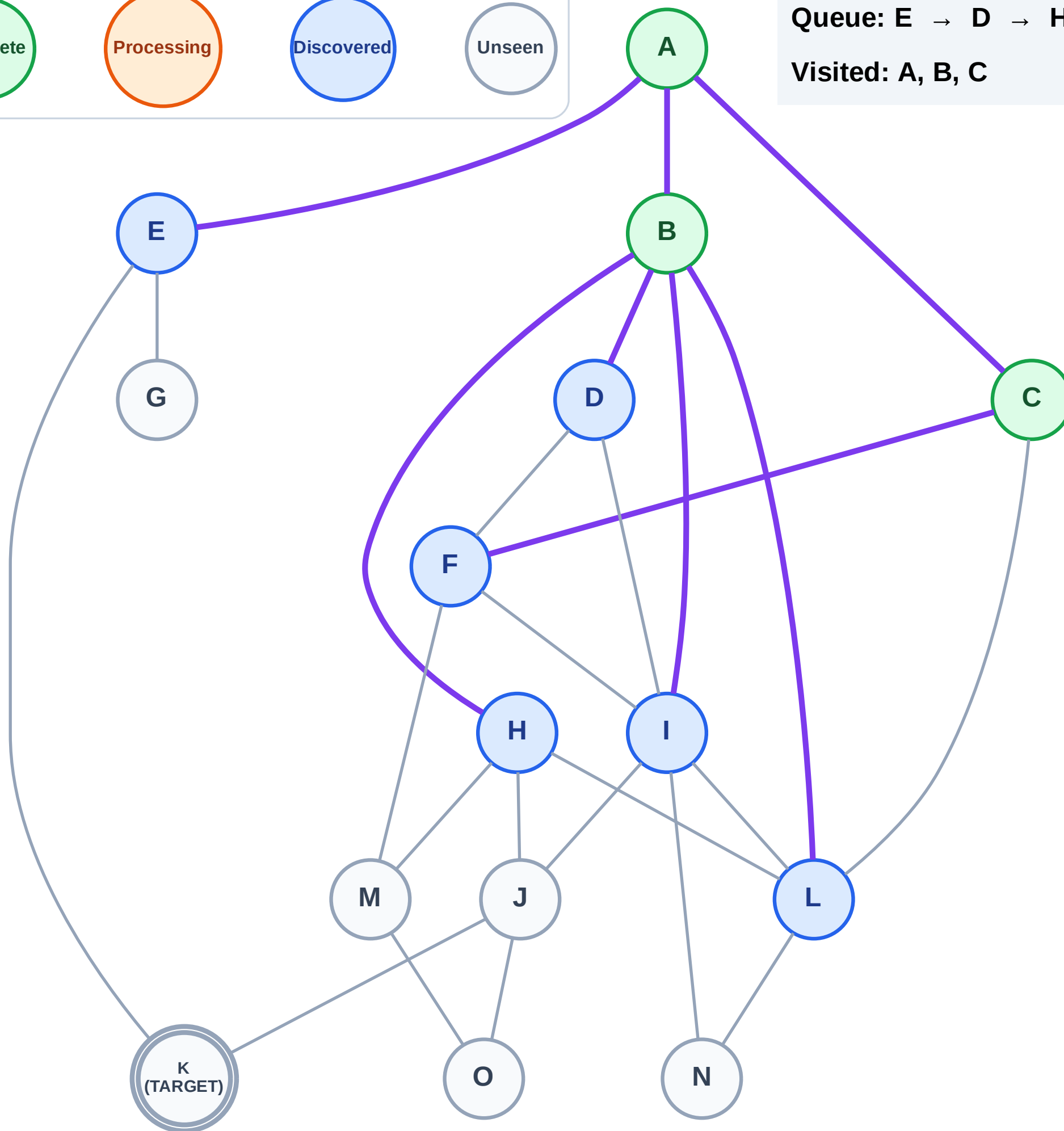
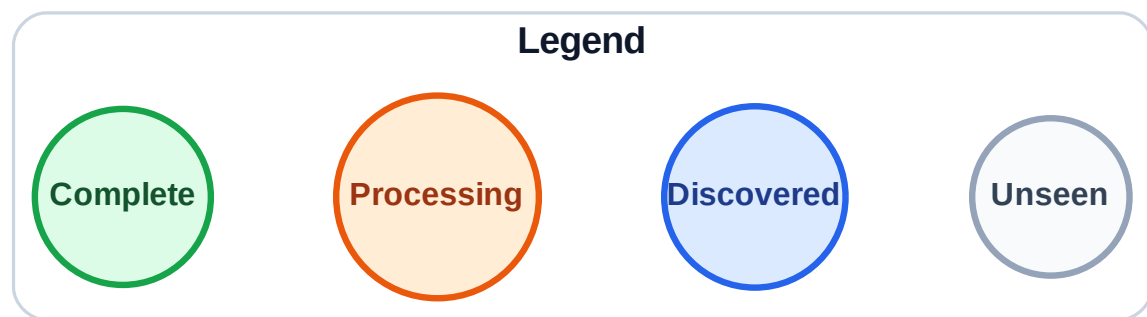
Visited: A, B



BFS Traversal

Step 7: Discover F from C

Queue: E → D → H → I → L → F
Visited: A, B, C



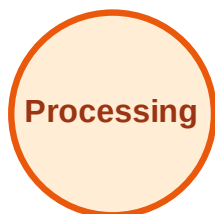
BFS Traversal

Step 8: Process node E

Legend



Complete



Processing



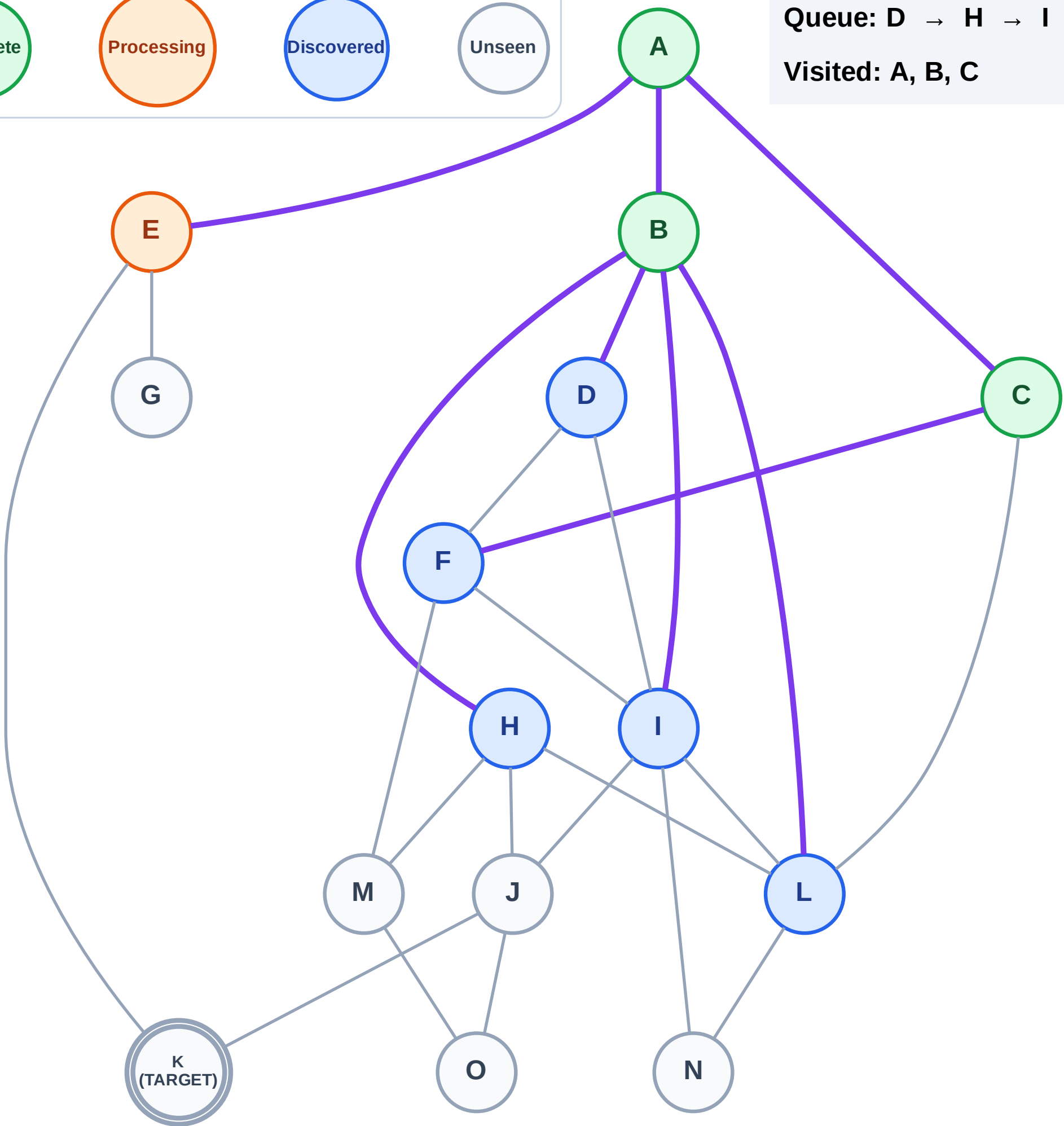
Discovered



Unseen

Queue: D → H → I → L → F

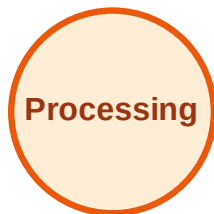
Visited: A, B, C



BFS Traversal

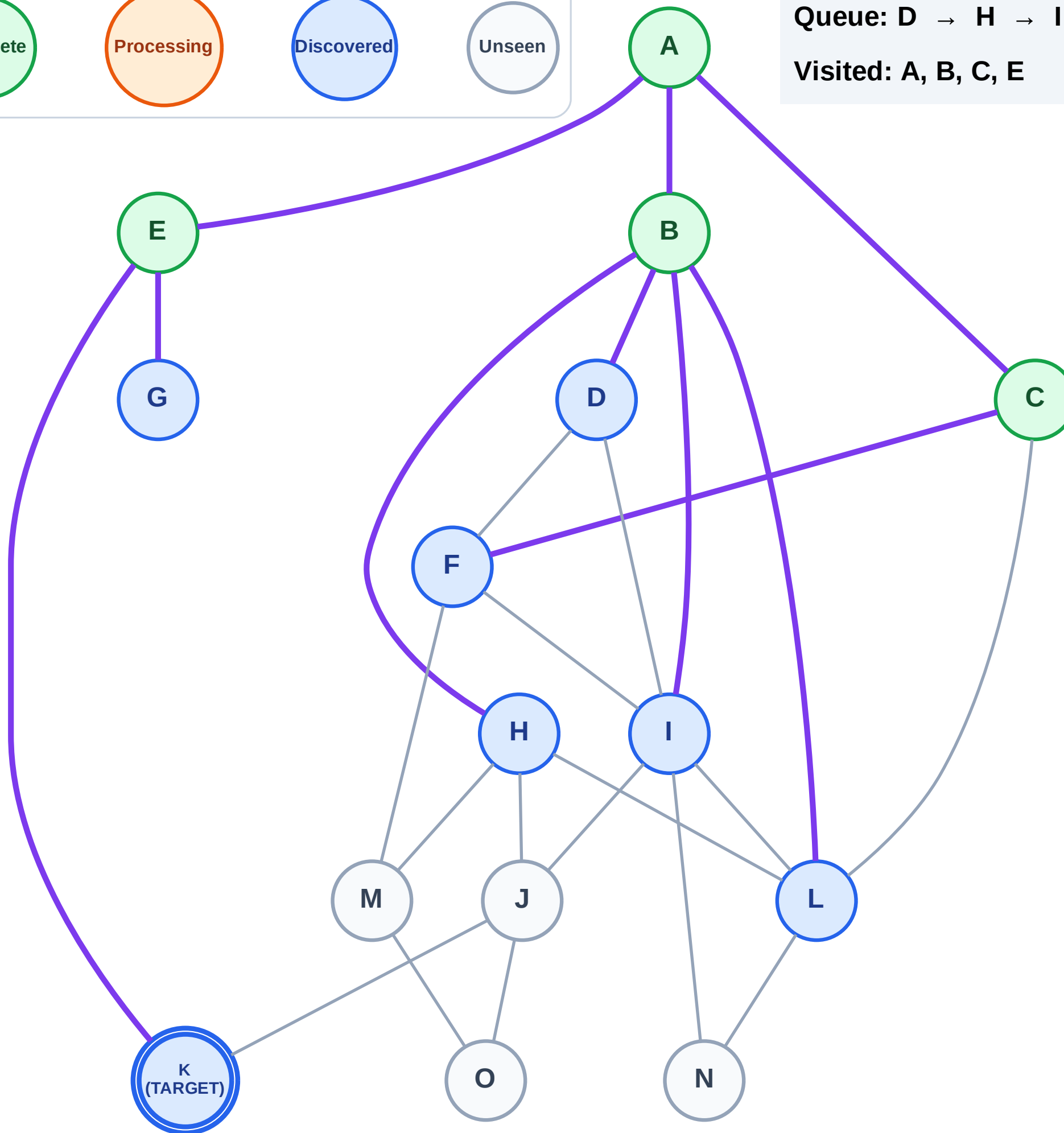
Step 9: Discover G, K from E

Legend



Queue: D → H → I → L → F → G → K

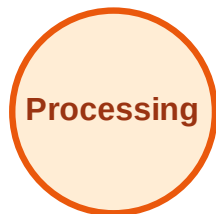
Visited: A, B, C, E



BFS Traversal

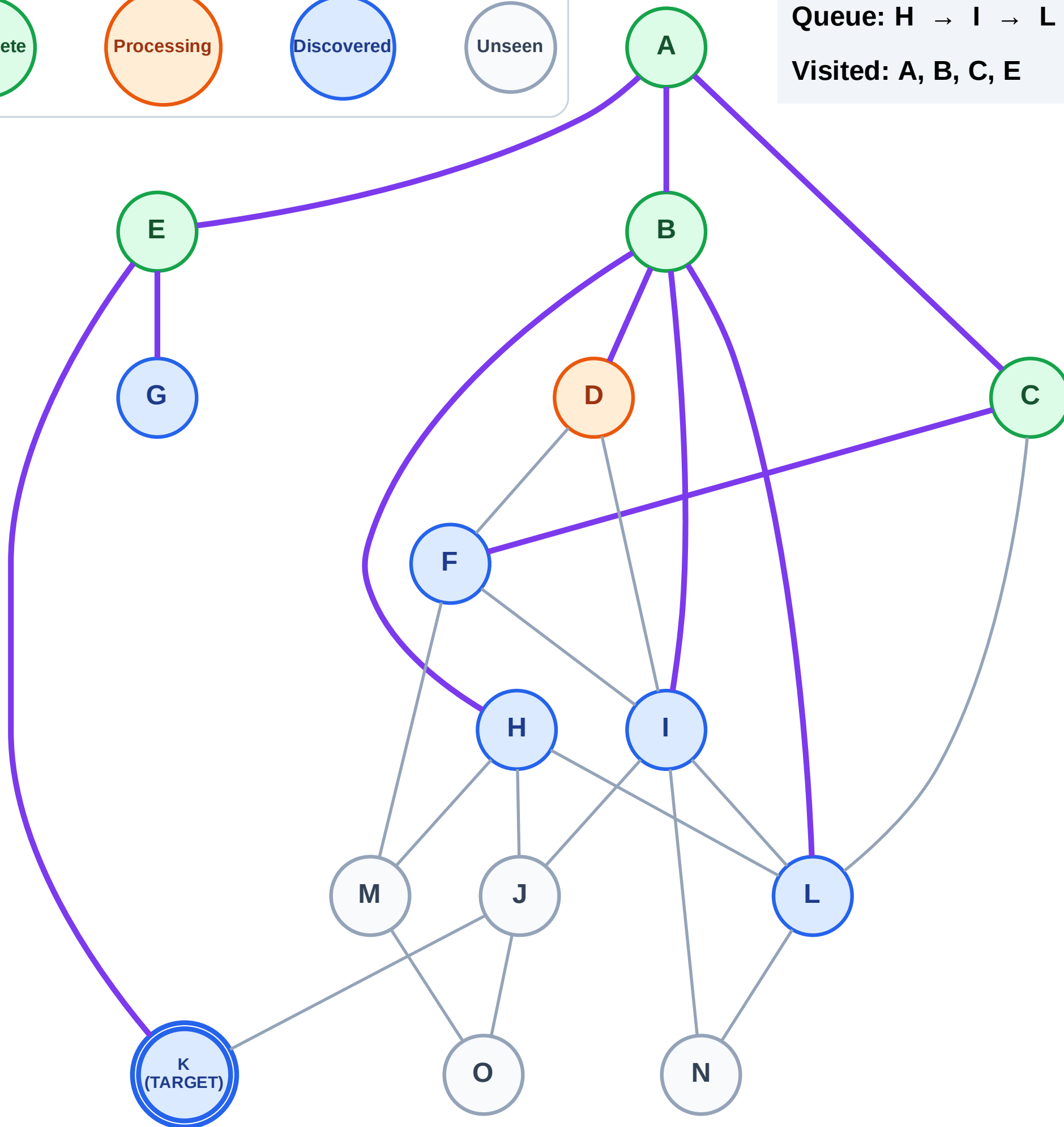
Step 10: Process node D

Legend



Queue: H → I → L → F → G → K

Visited: A, B, C, E



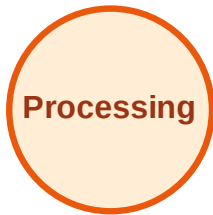
BFS Traversal

Step 11: No new neighbors from D

Legend



Complete



Processing



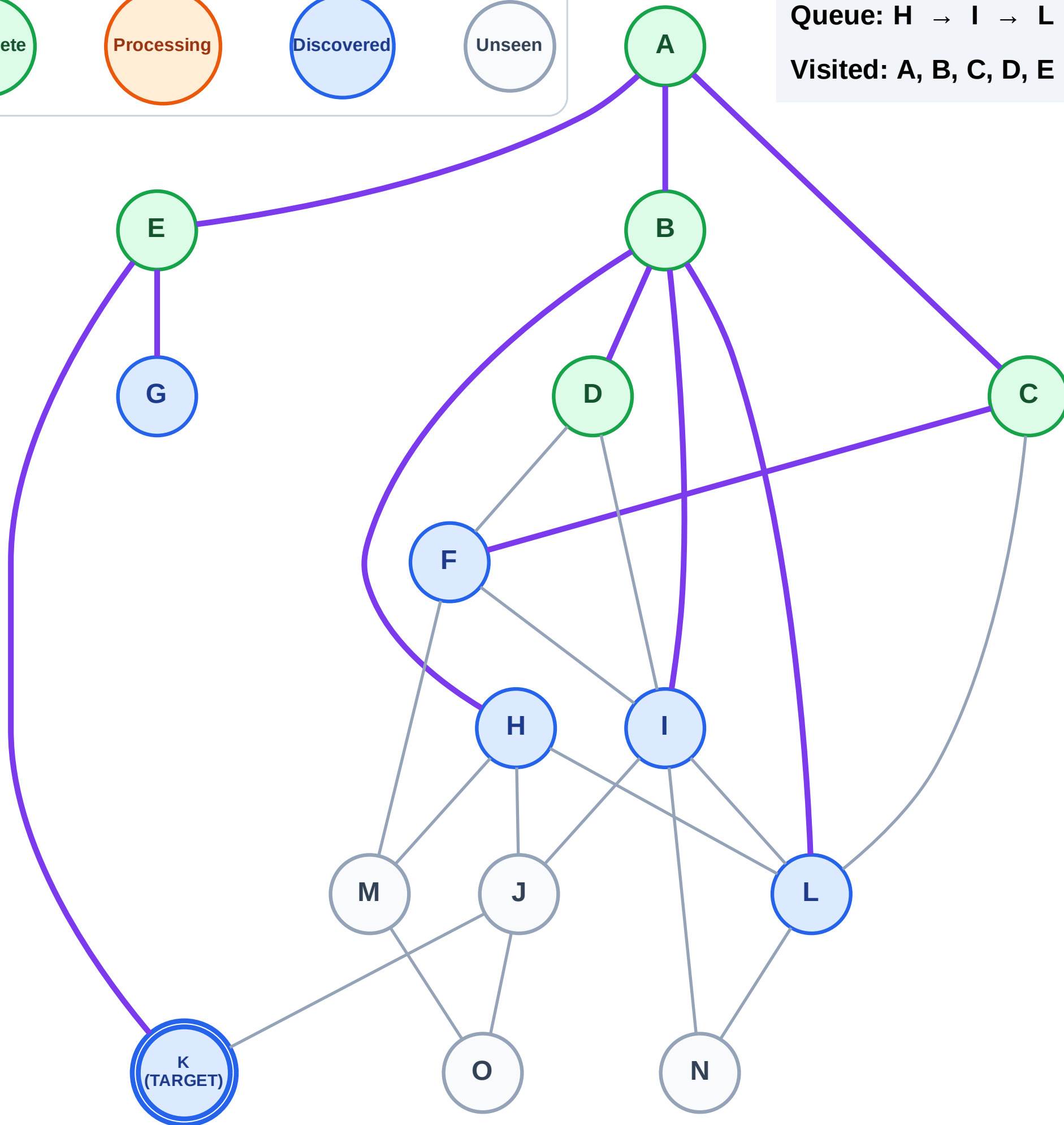
Discovered



Unseen

Queue: H → I → L → F → G → K

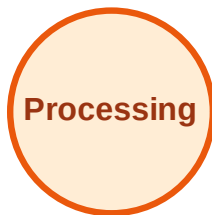
Visited: A, B, C, D, E



BFS Traversal

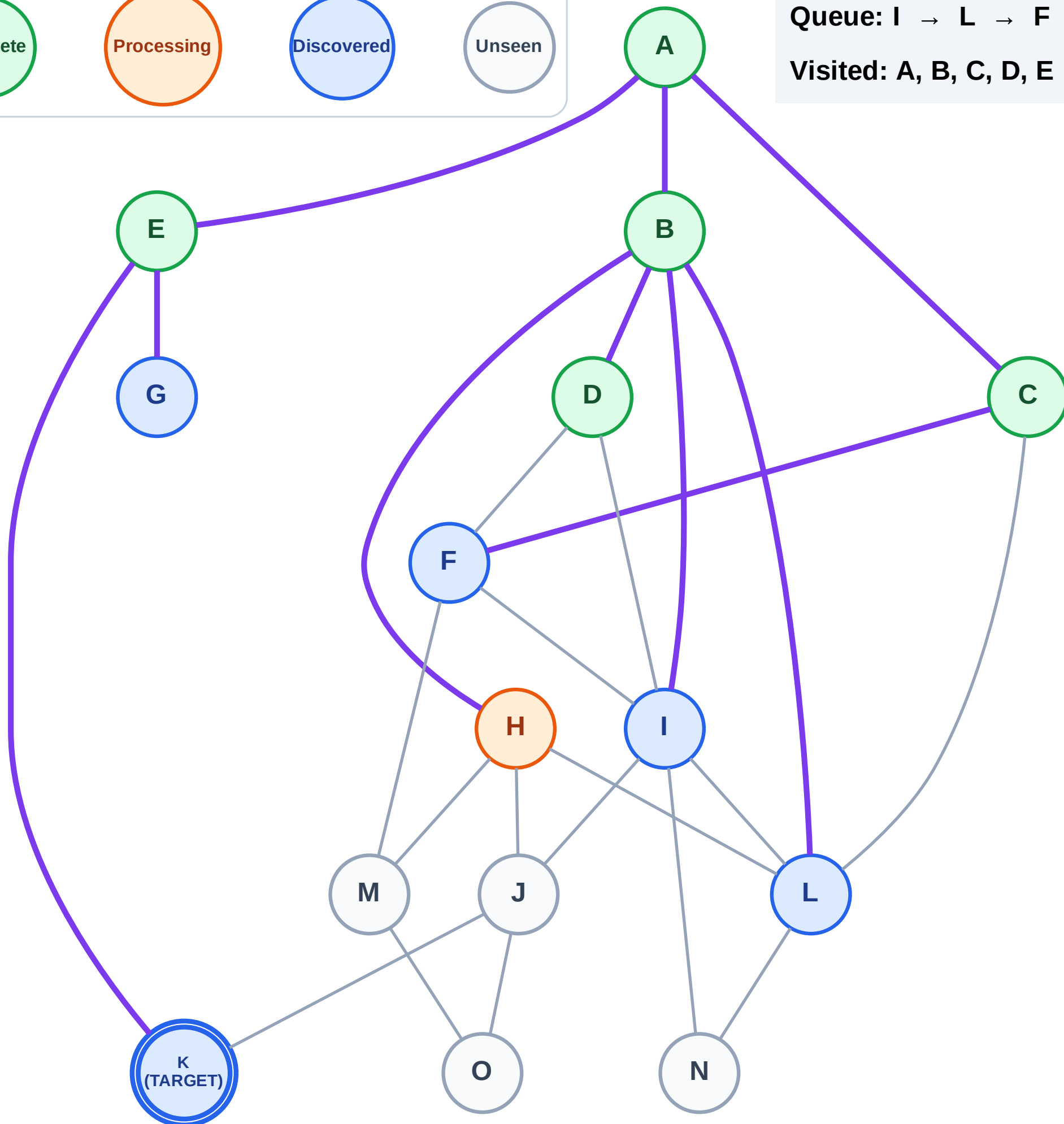
Step 12: Process node H

Legend



Queue: I → L → F → G → K

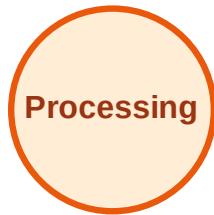
Visited: A, B, C, D, E



BFS Traversal

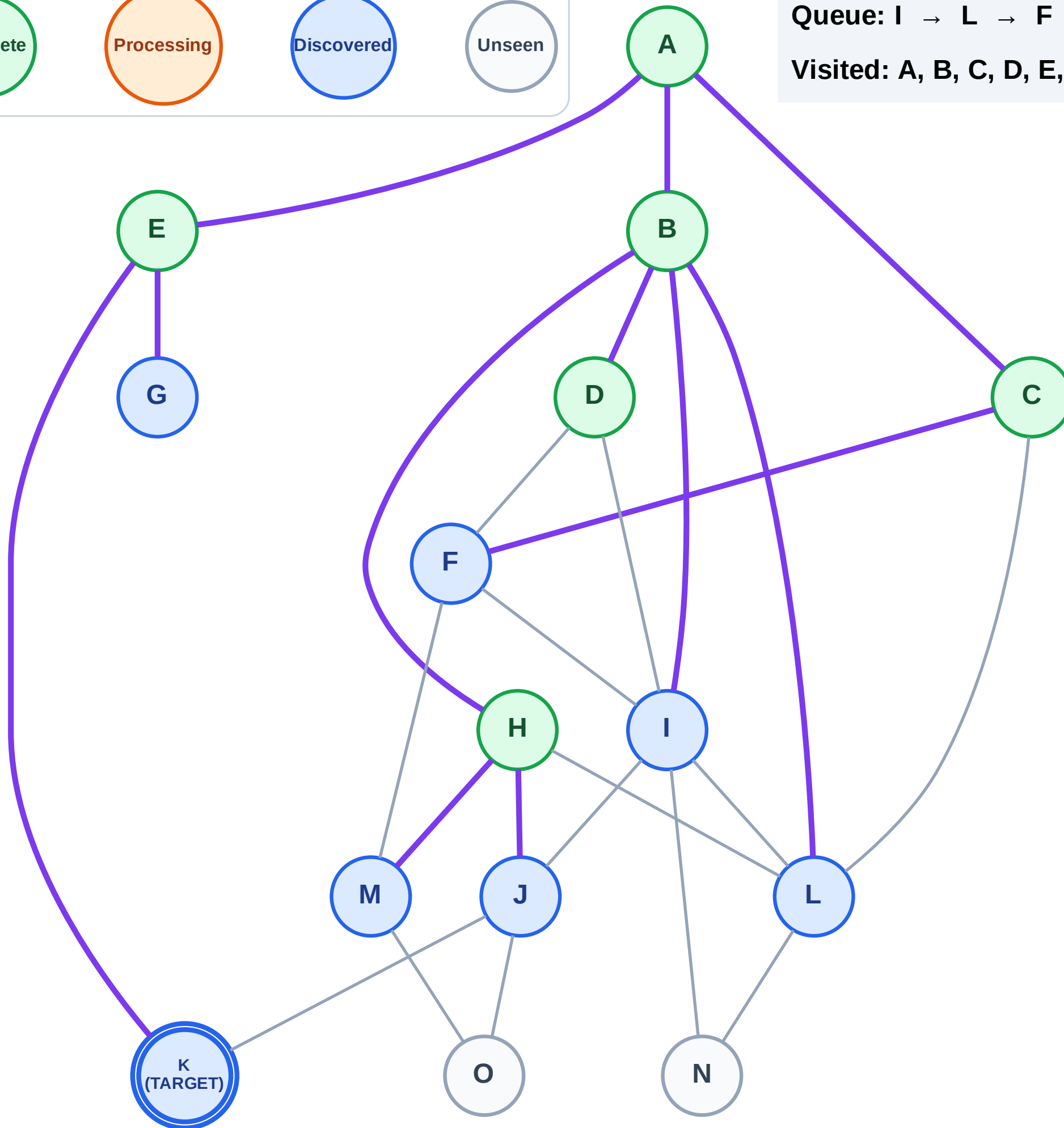
Step 13: Discover J, M from H

Legend



Queue: I → L → F → G → K → J → M

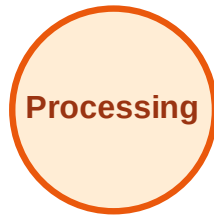
Visited: A, B, C, D, E, H



BFS Traversal

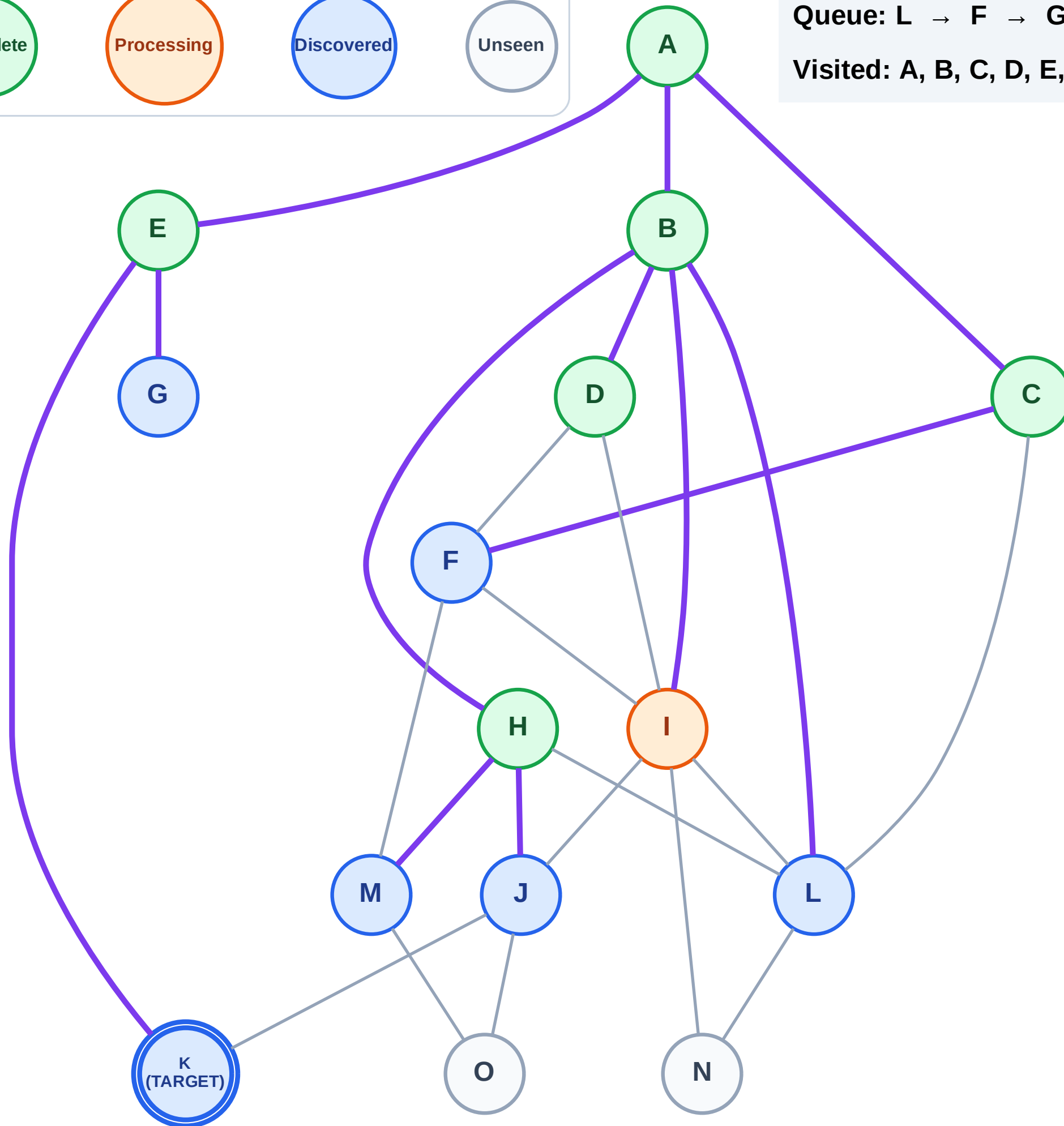
Step 14: Process node I

Legend



Queue: L → F → G → K → J → M

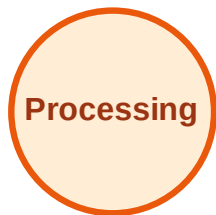
Visited: A, B, C, D, E, H



BFS Traversal

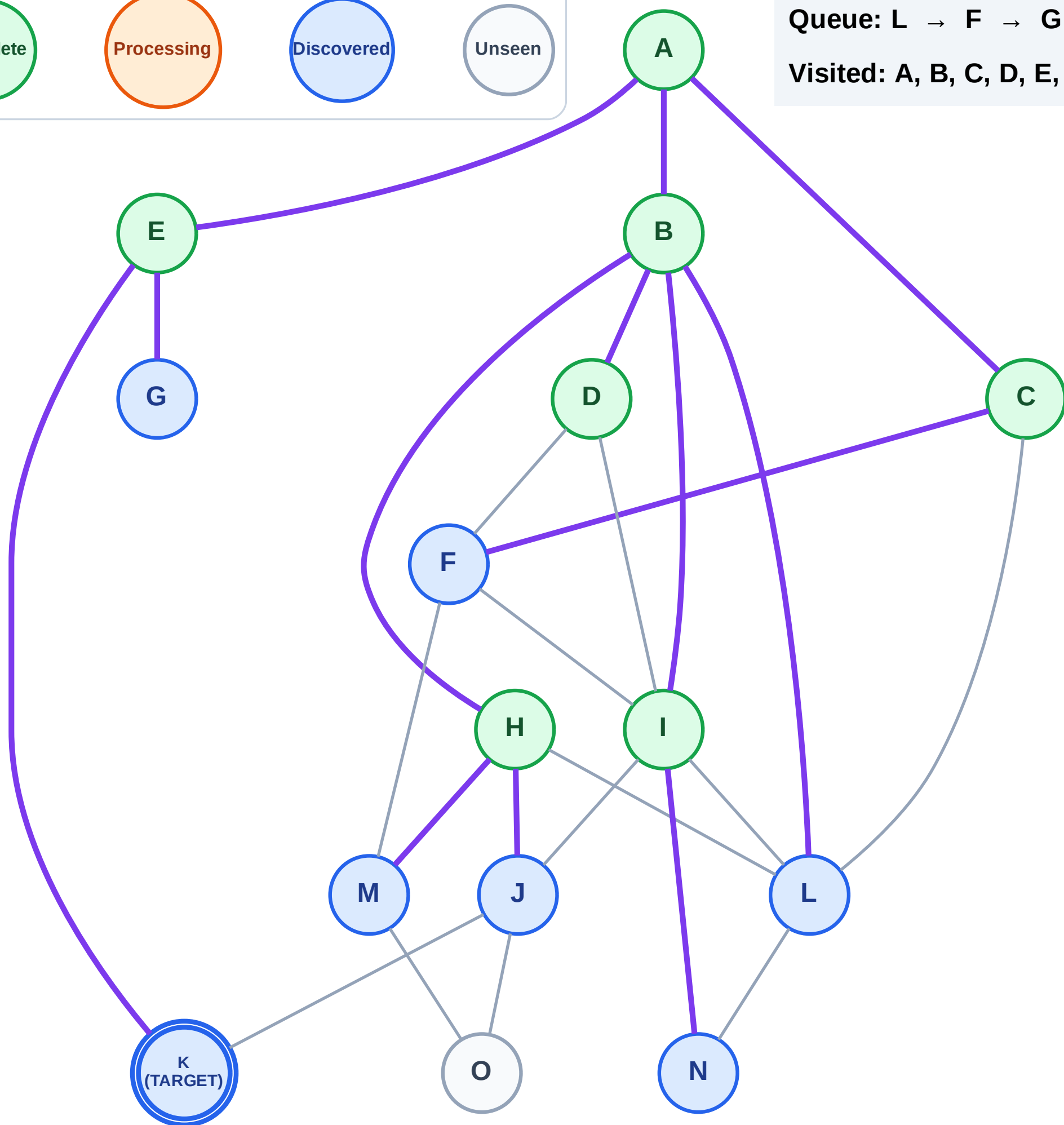
Step 15: Discover N from I

Legend



Queue: L → F → G → K → J → M → N

Visited: A, B, C, D, E, H, I



BFS Traversal

Step 16: Process node L

Legend

Complete

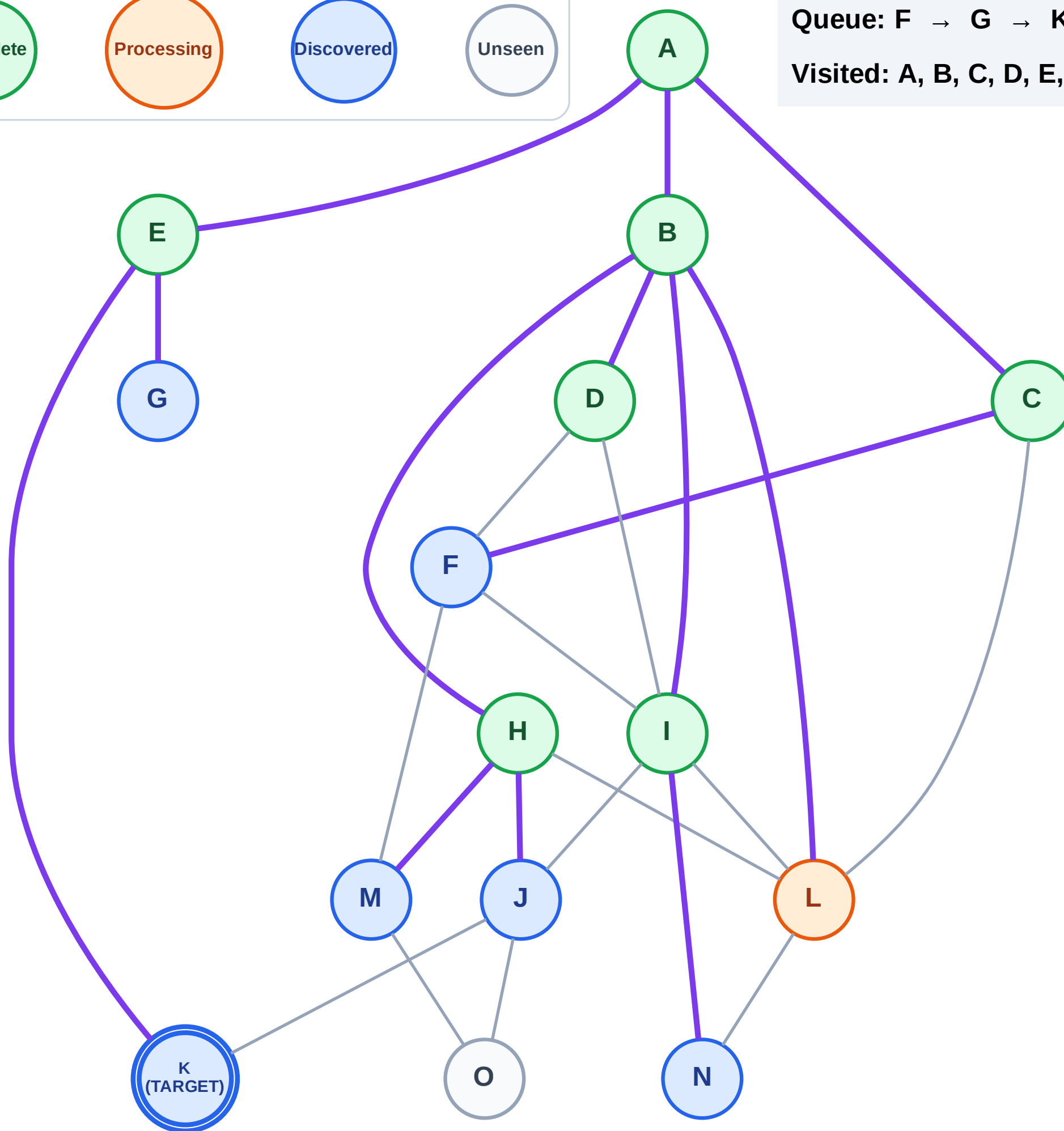
Processing

Discovered

Unseen

Queue: F → G → K → J → M → N

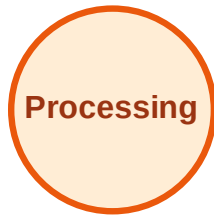
Visited: A, B, C, D, E, H, I



BFS Traversal

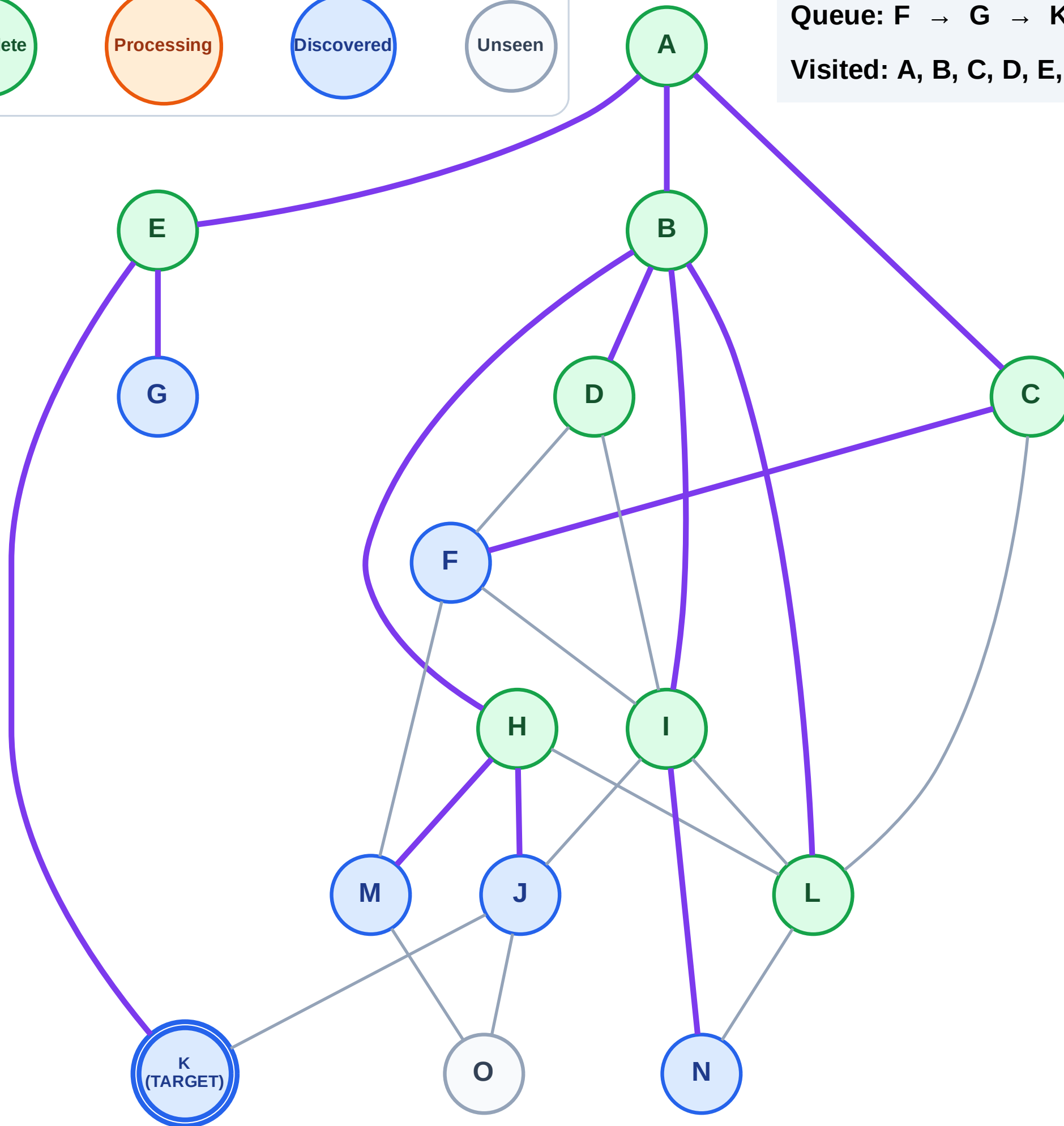
Step 17: No new neighbors from L

Legend



Queue: F → G → K → J → M → N

Visited: A, B, C, D, E, H, I, L



BFS Traversal

Step 18: Process node F

Legend

Complete

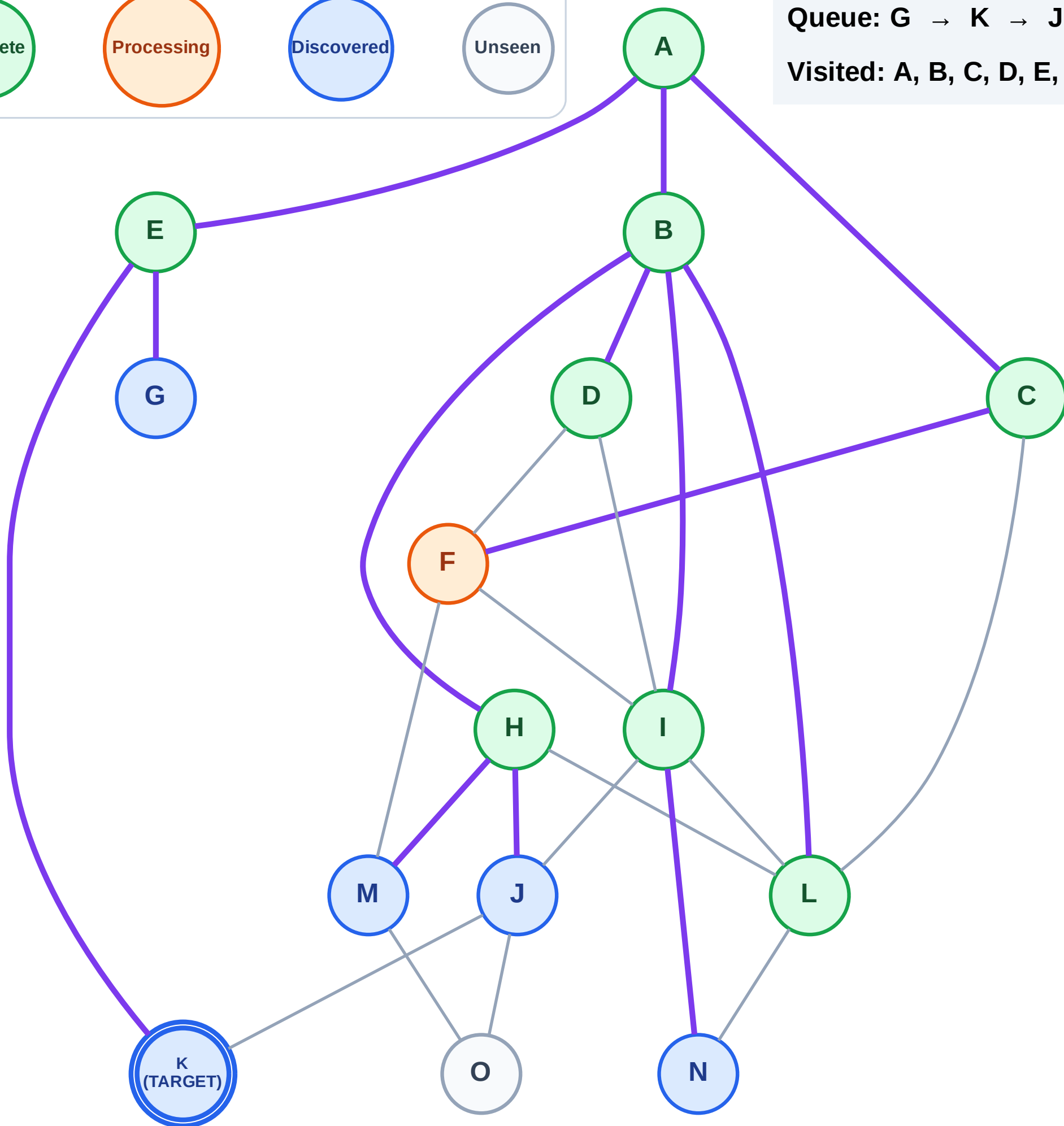
Processing

Discovered

Unseen

Queue: G → K → J → M → N

Visited: A, B, C, D, E, H, I, L



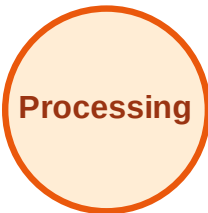
BFS Traversal

Step 19: No new neighbors from F

Legend



Complete



Processing



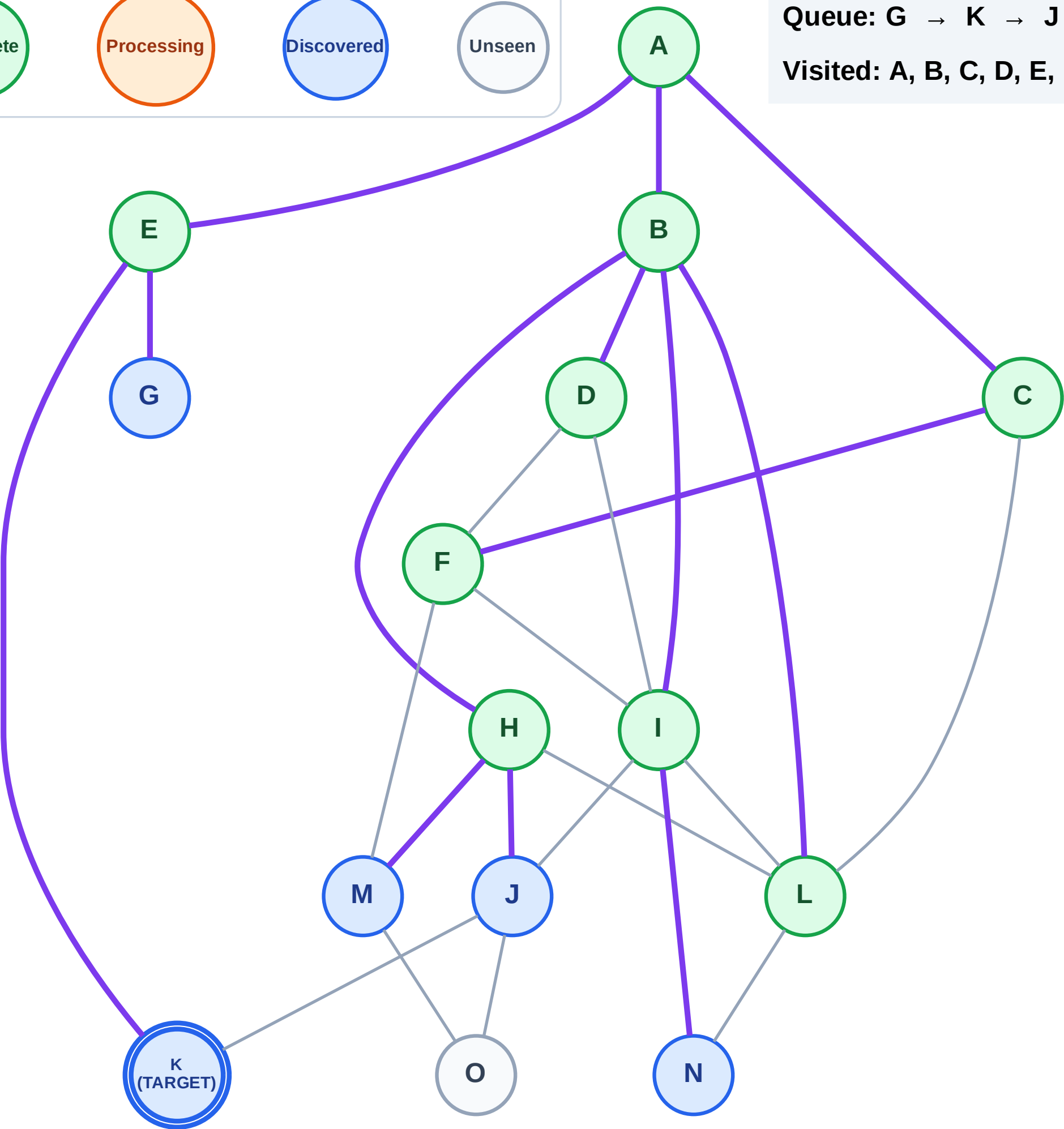
Discovered



Unseen

Queue: G → K → J → M → N

Visited: A, B, C, D, E, F, H, I, L



BFS Traversal

Step 20: Process node G

Legend

Complete

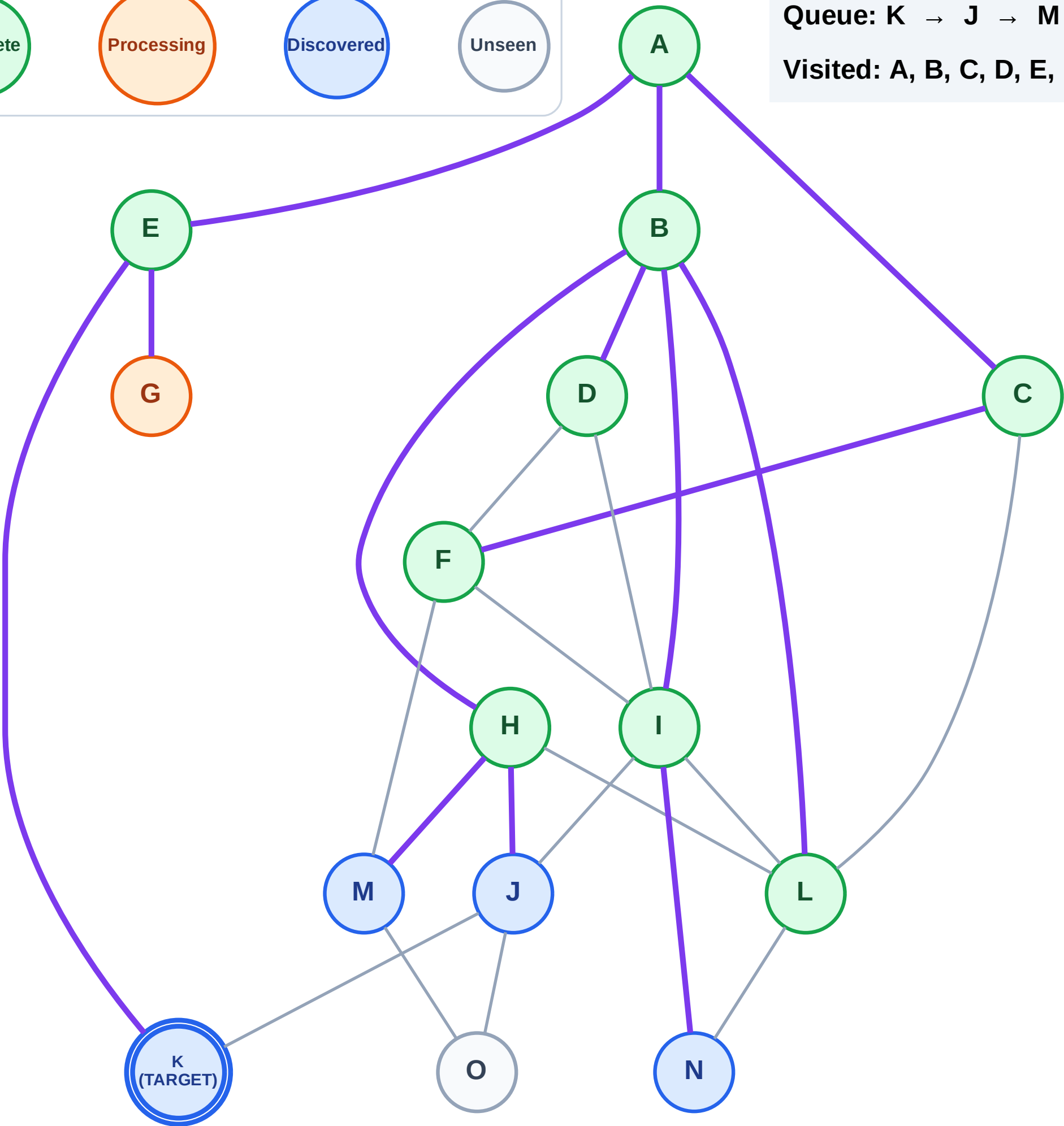
Processing

Discovered

Unseen

Queue: K → J → M → N

Visited: A, B, C, D, E, F, H, I, L



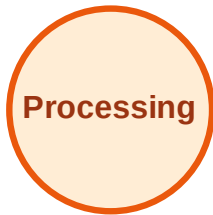
BFS Traversal

Step 21: No new neighbors from G

Legend



Complete



Processing



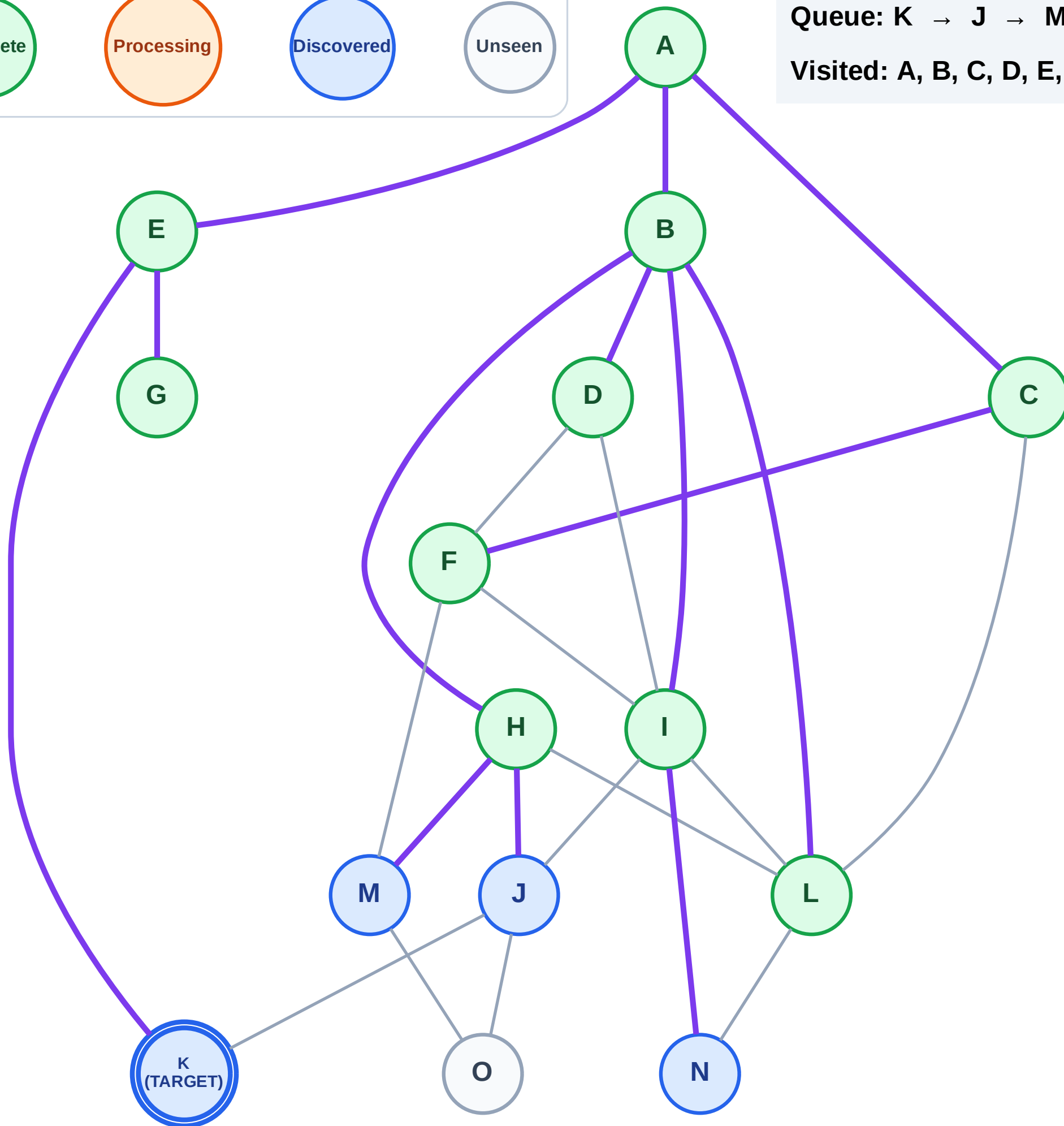
Discovered



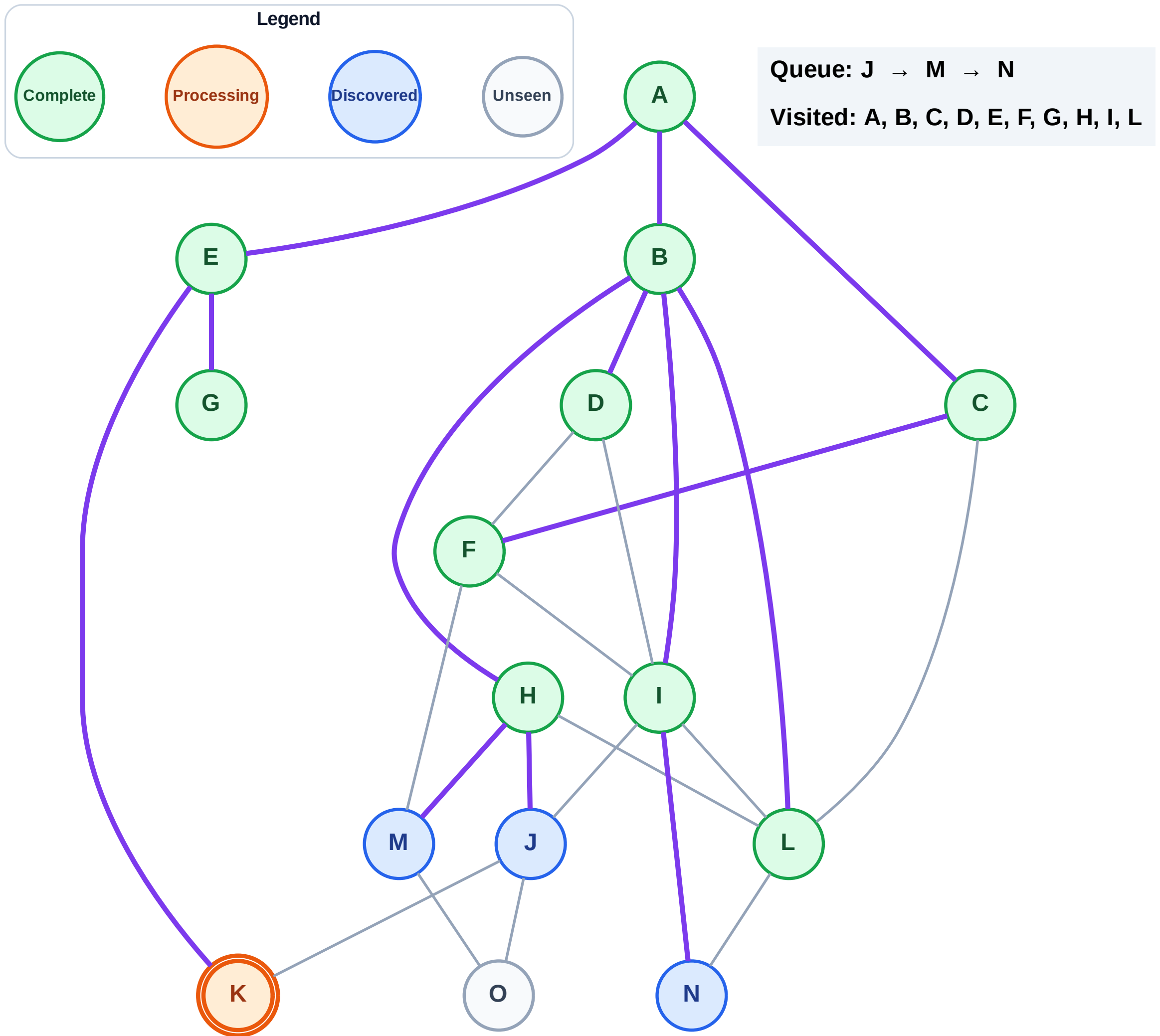
Unseen

Queue: K → J → M → N

Visited: A, B, C, D, E, F, G, H, I, L



Step 22: Process node K



BFS Traversal

Step 23: TARGET NODE K REACHED! Path (2 hops): A → E → K

Legend

Complete

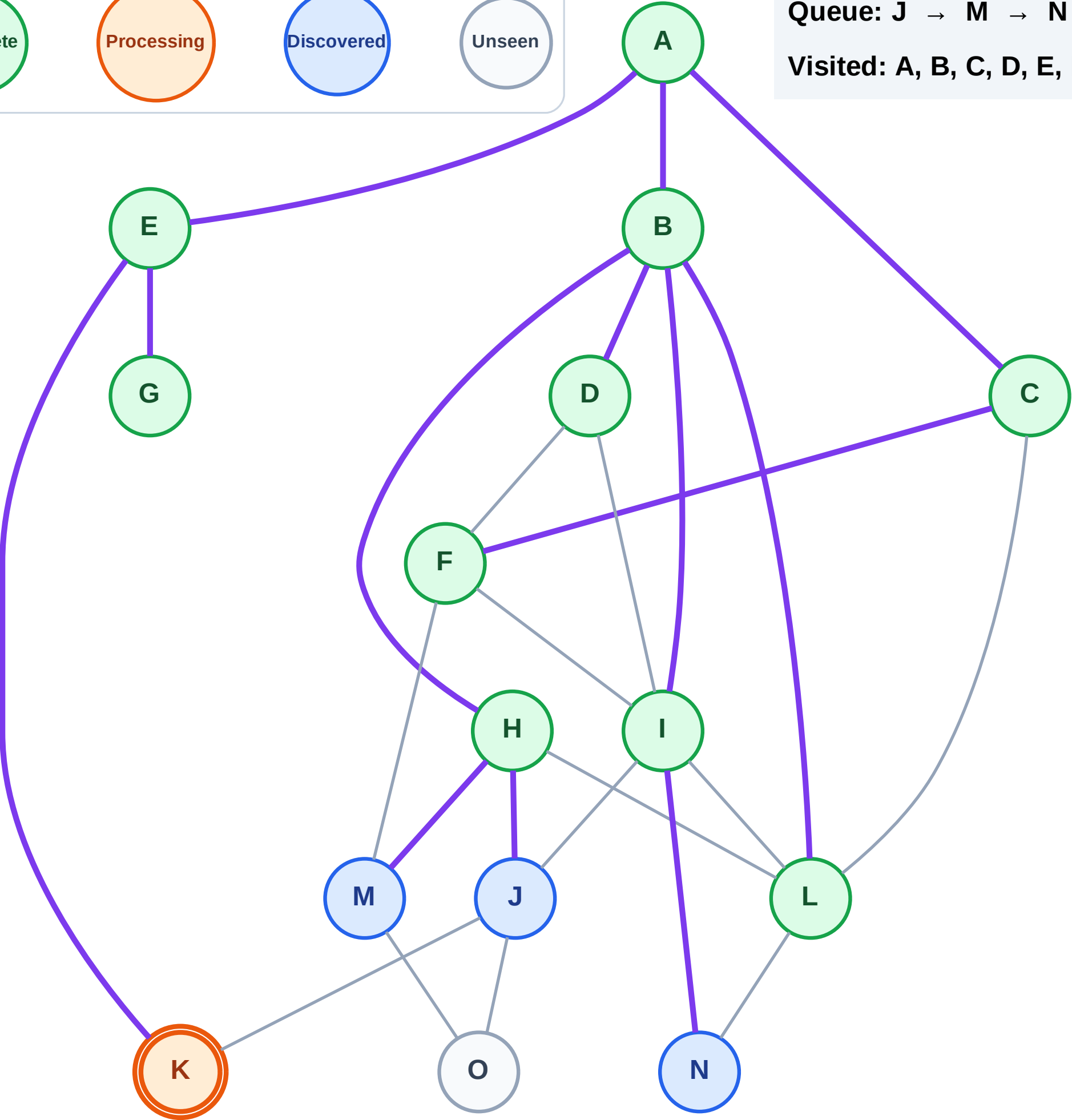
Processing

Discovered

Unseen

Queue: J → M → N

Visited: A, B, C, D, E, F, G, H, I, K, L



DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend

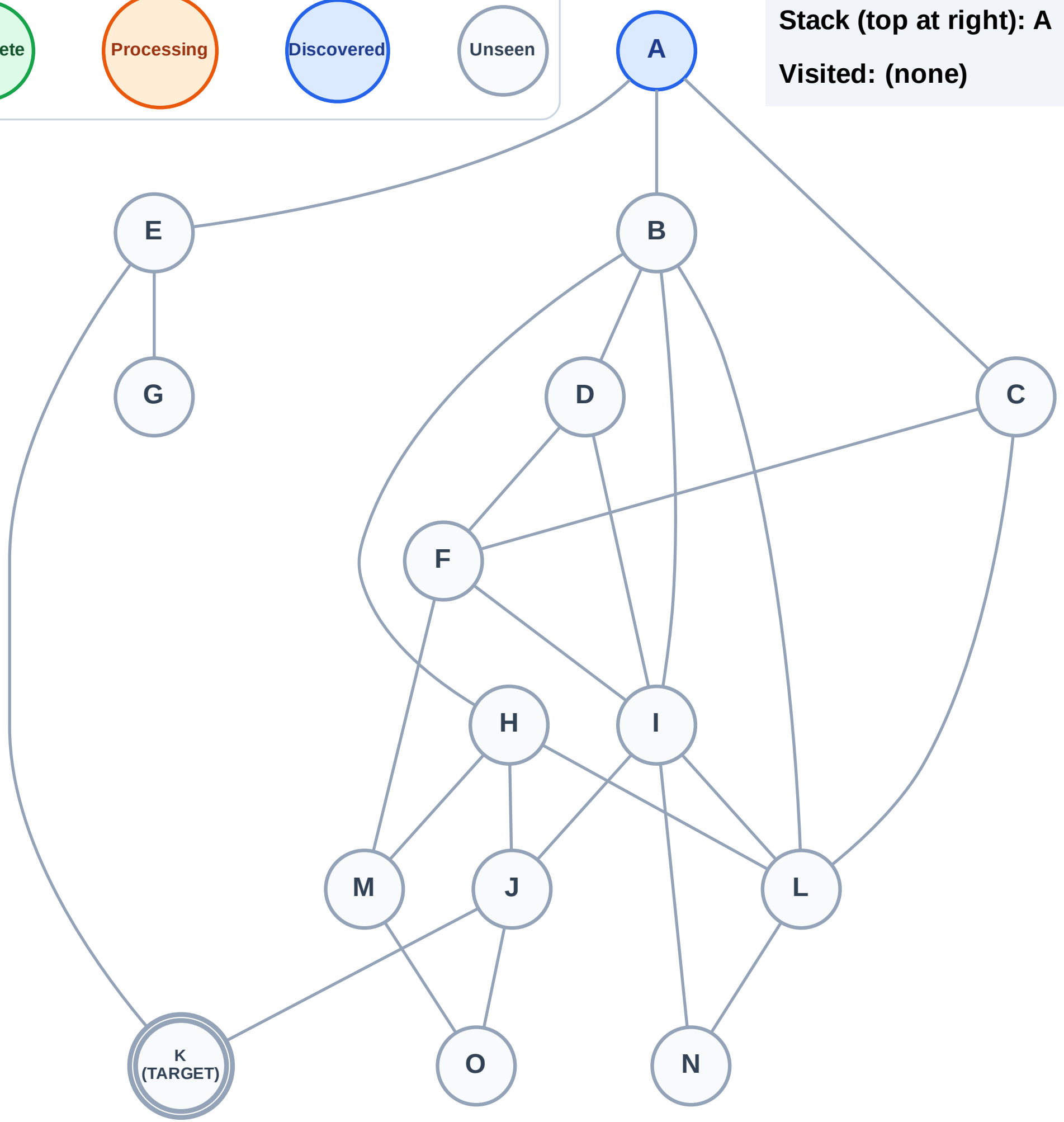
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

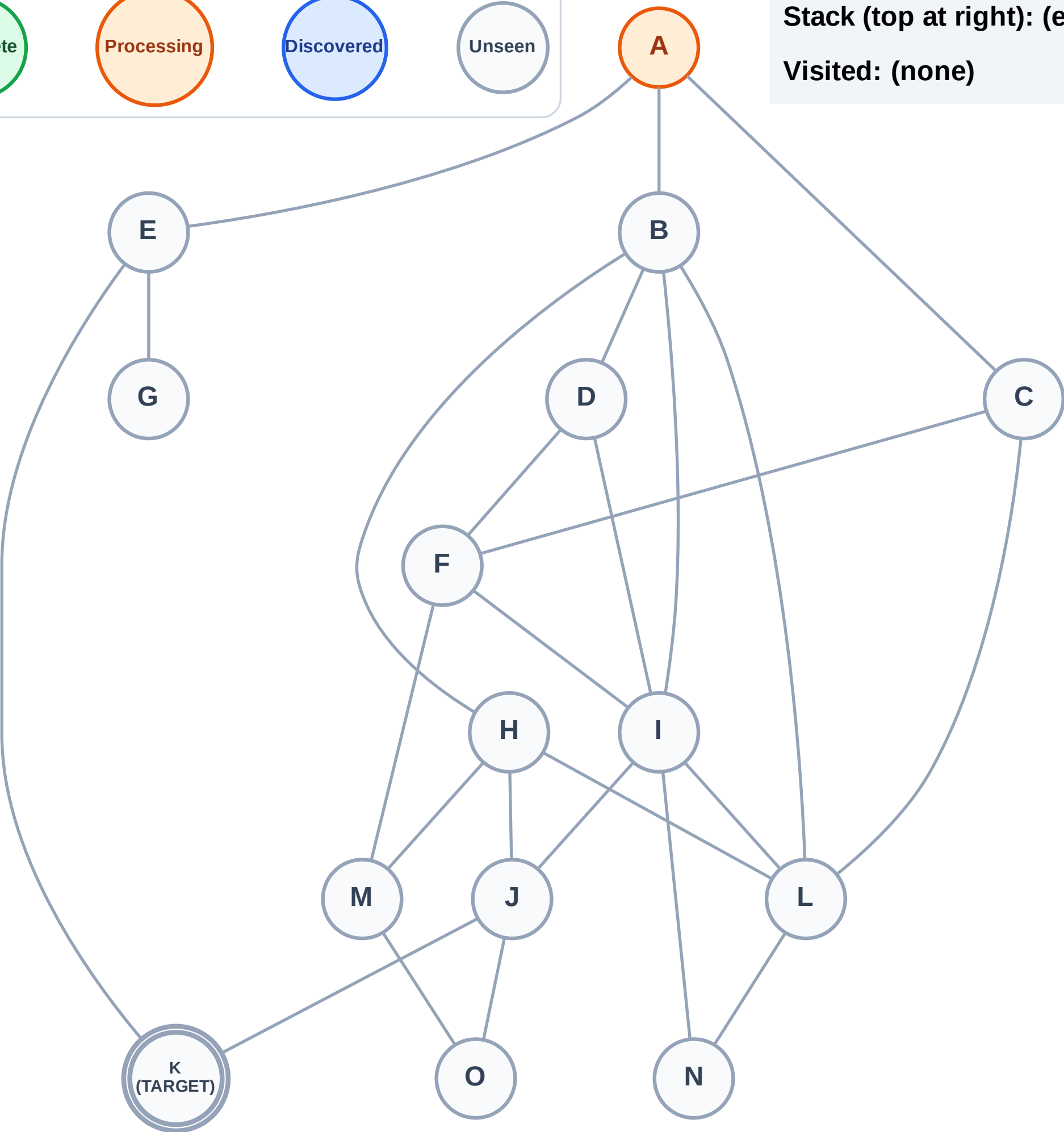
Complete

Processing

Discovered

Unseen

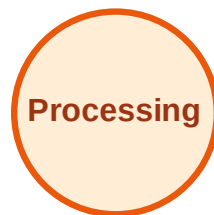
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

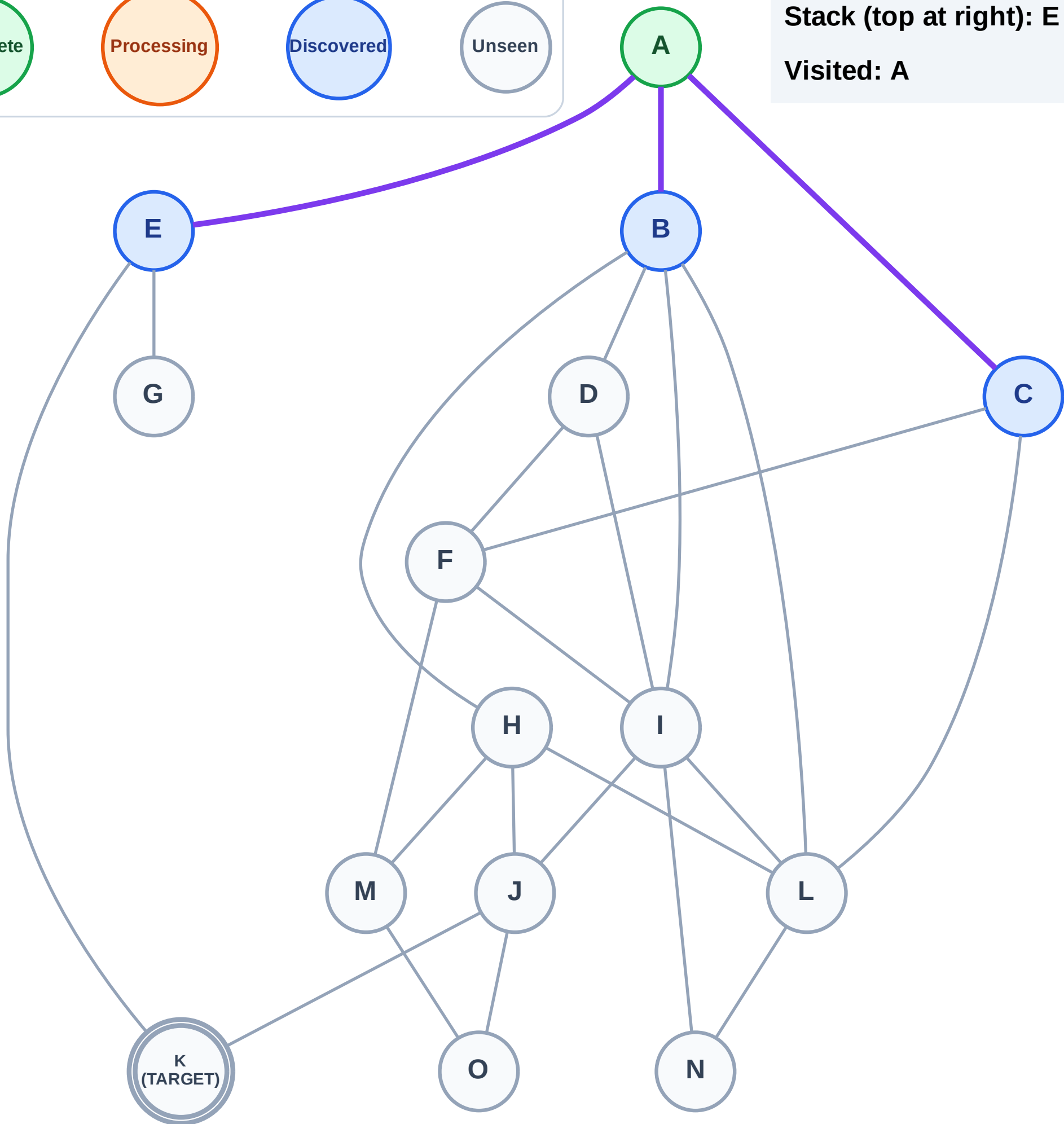
Step 3: Discover E, C, B from A

Legend



Stack (top at right): E | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

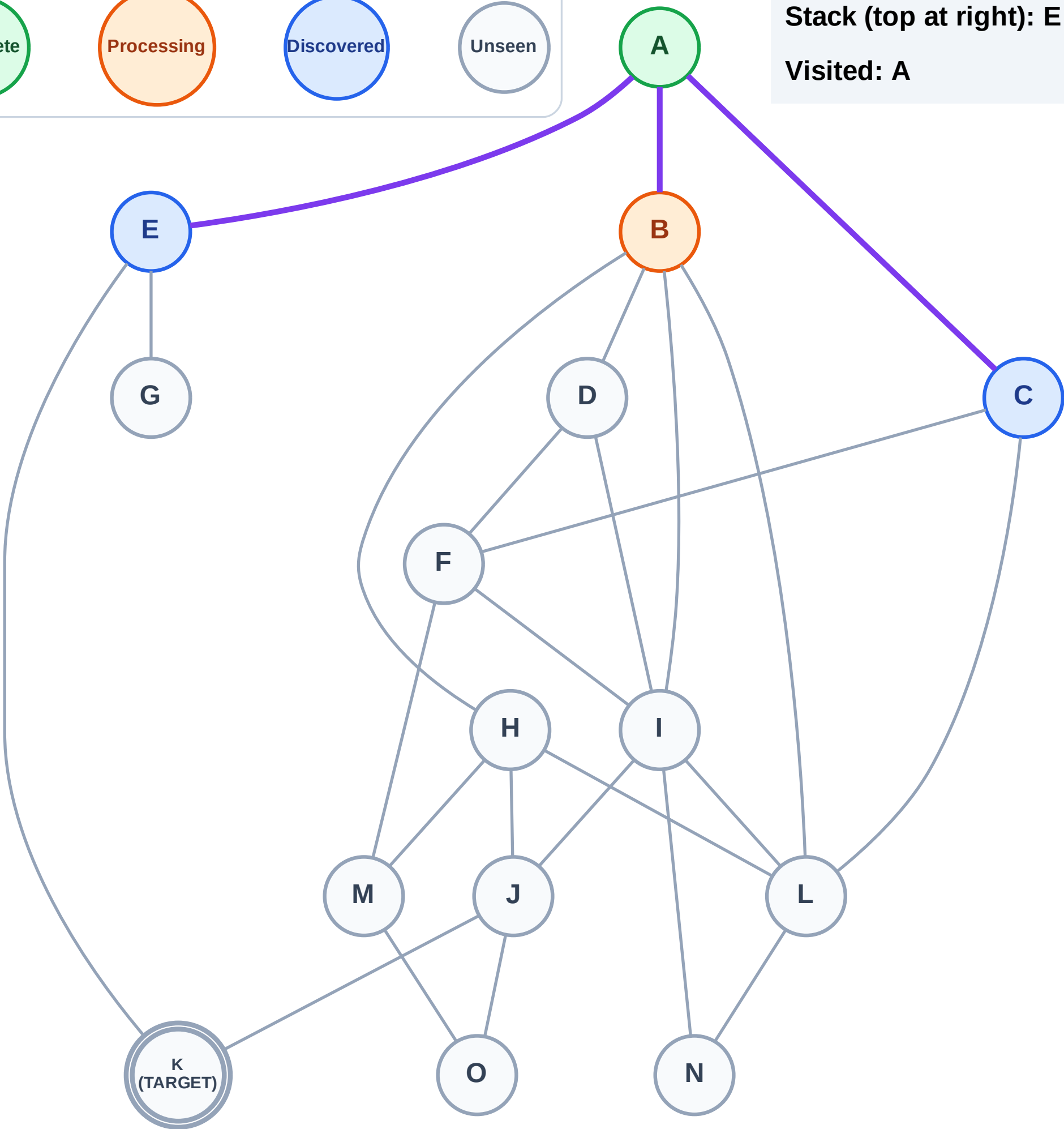
Complete

Processing

Discovered

Unseen

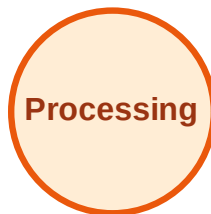
Stack (top at right): E | C
Visited: A



DFS Traversal

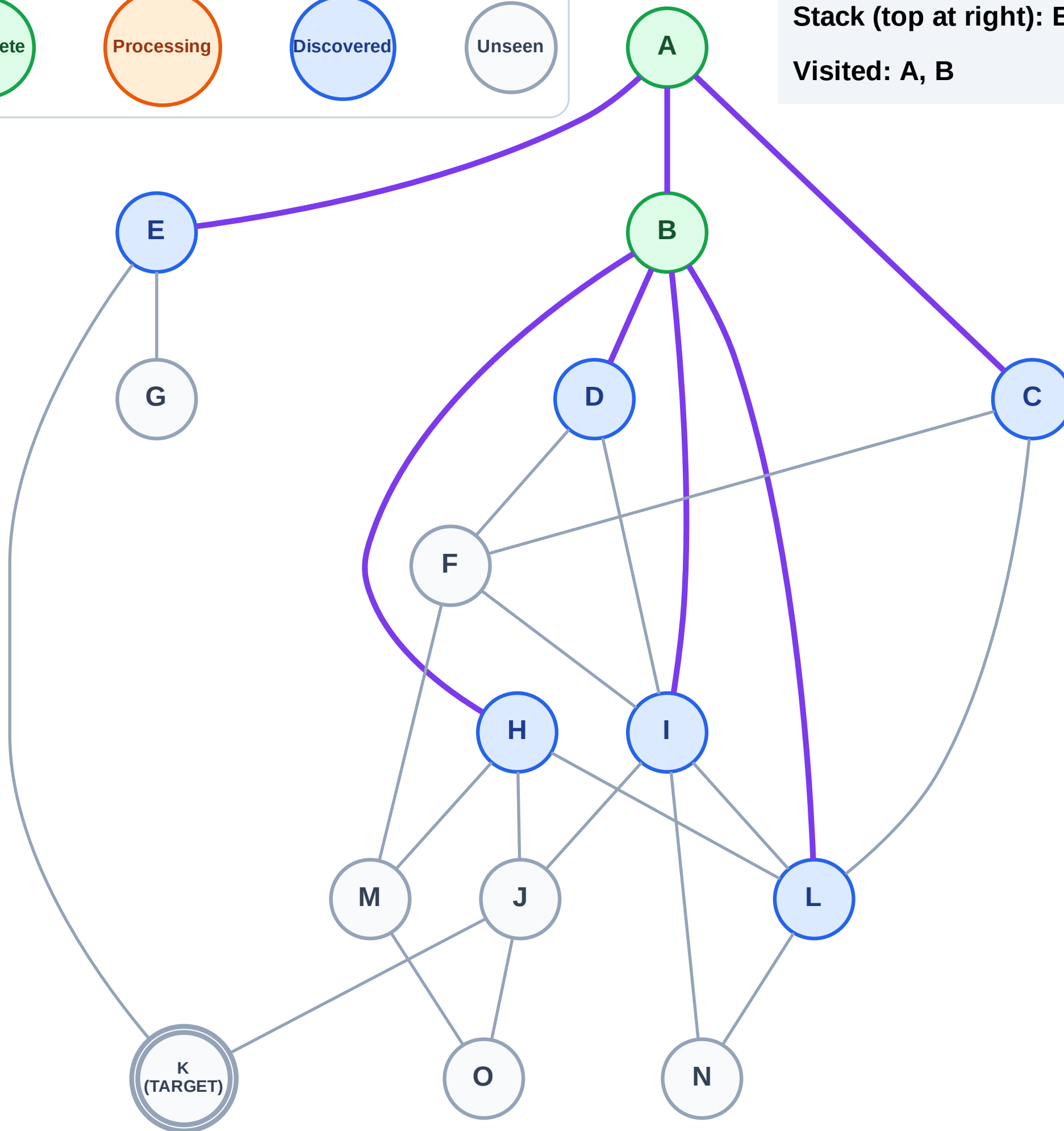
Step 5: Discover L, I, H, D from B

Legend



Stack (top at right): E | C | L | I | H | D

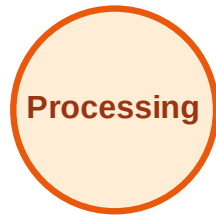
Visited: A, B



DFS Traversal

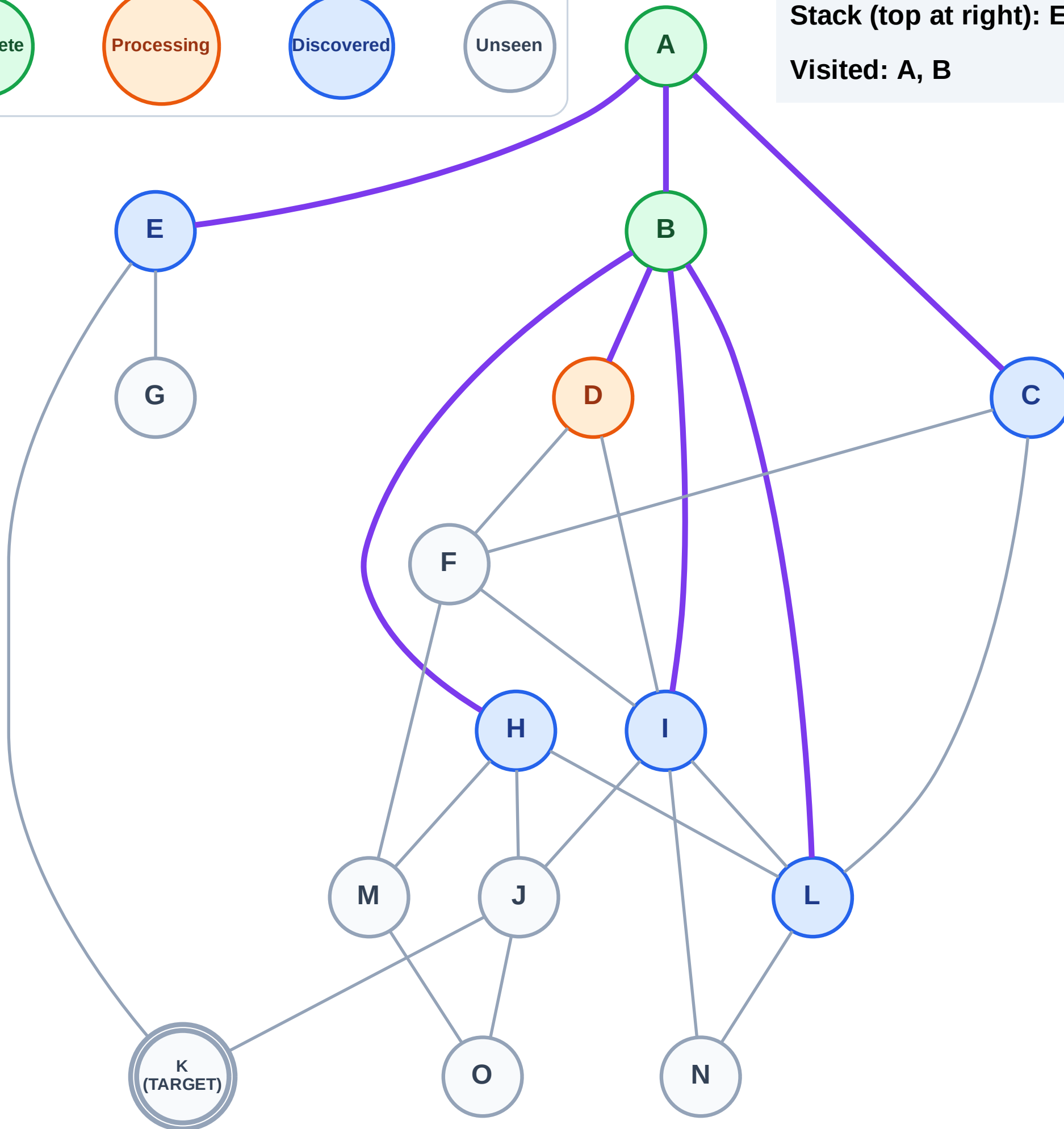
Step 6: Process node D

Legend



Stack (top at right): E | C | L | I | H

Visited: A, B



DFS Traversal

Step 7: Discover F from D

Legend

Complete

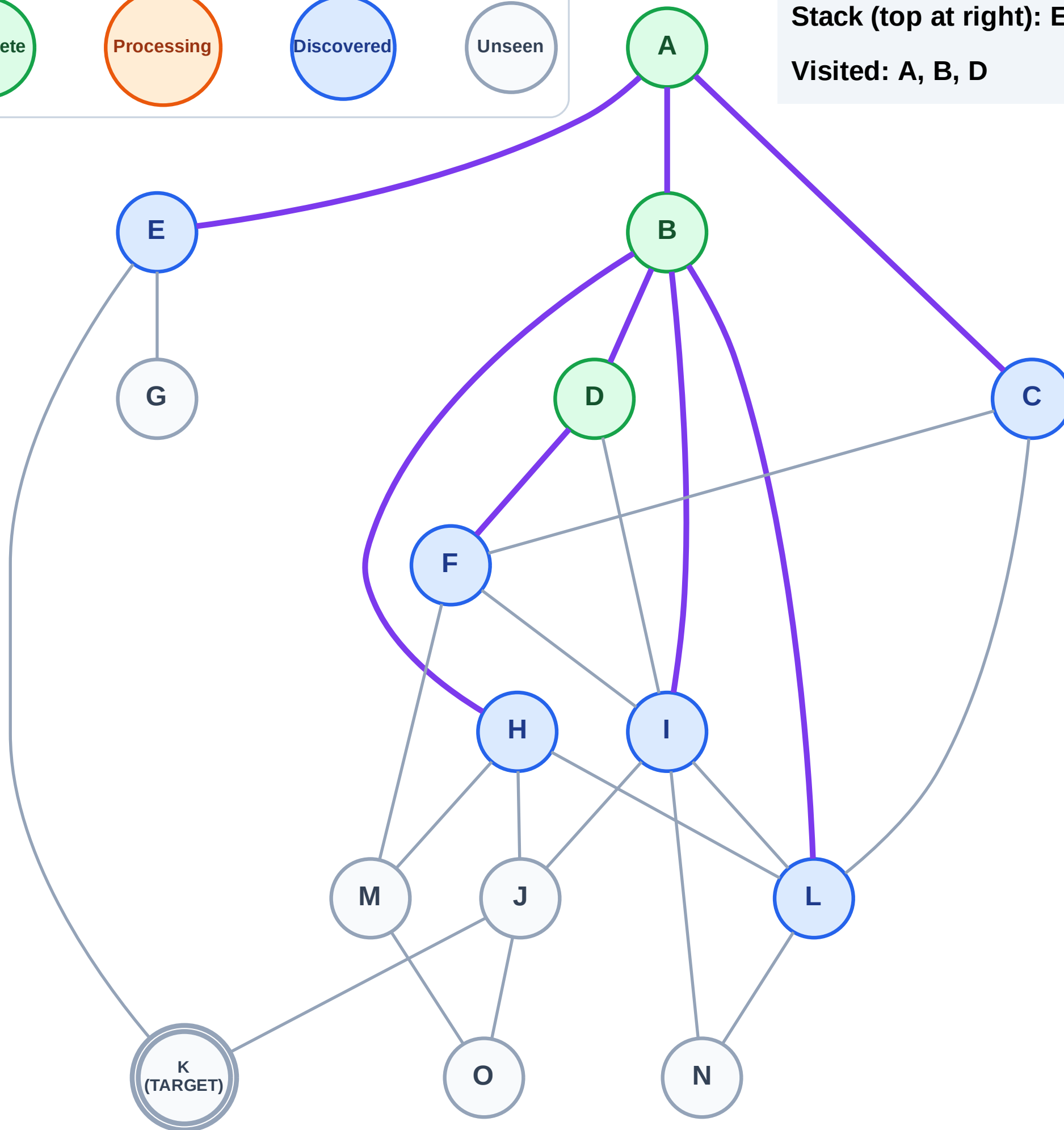
Processing

Discovered

Unseen

Stack (top at right): E | C | L | I | H | F

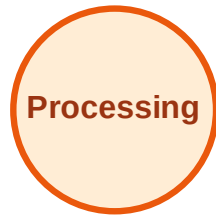
Visited: A, B, D



DFS Traversal

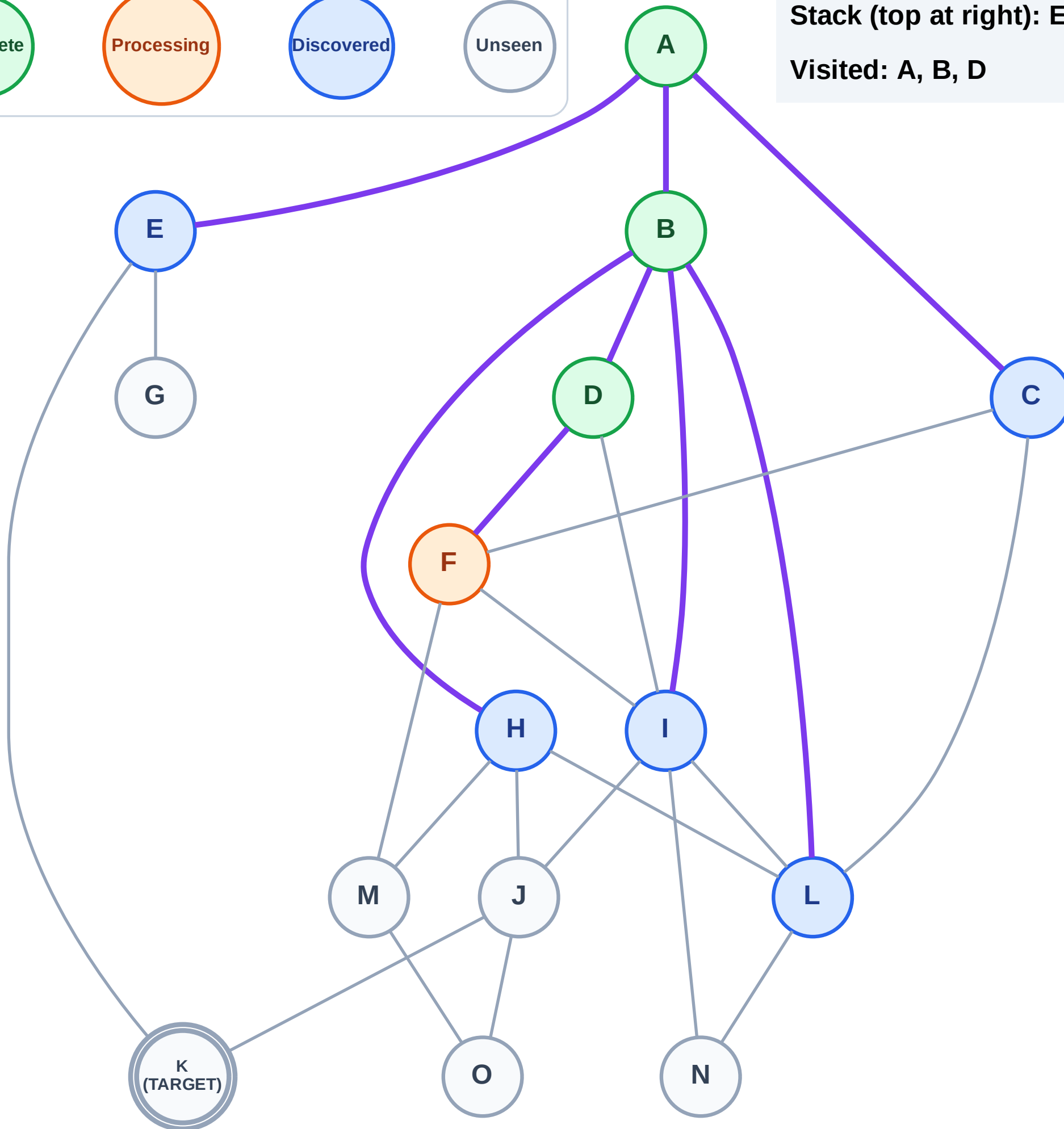
Step 8: Process node F

Legend



Stack (top at right): E | C | L | I | H

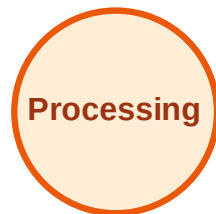
Visited: A, B, D



DFS Traversal

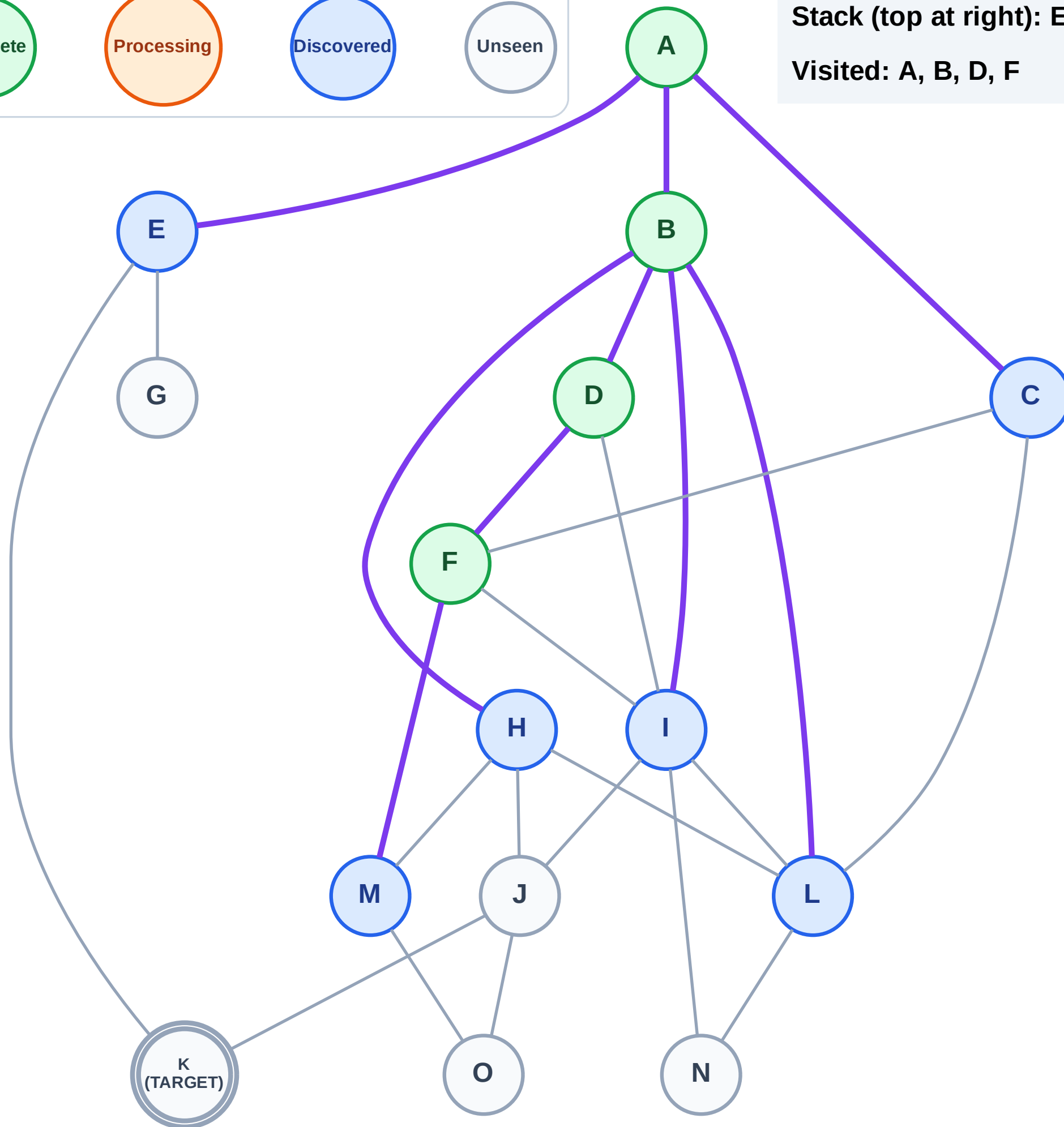
Step 9: Discover M from F

Legend



Stack (top at right): E | C | L | I | H | M

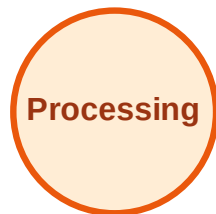
Visited: A, B, D, F



DFS Traversal

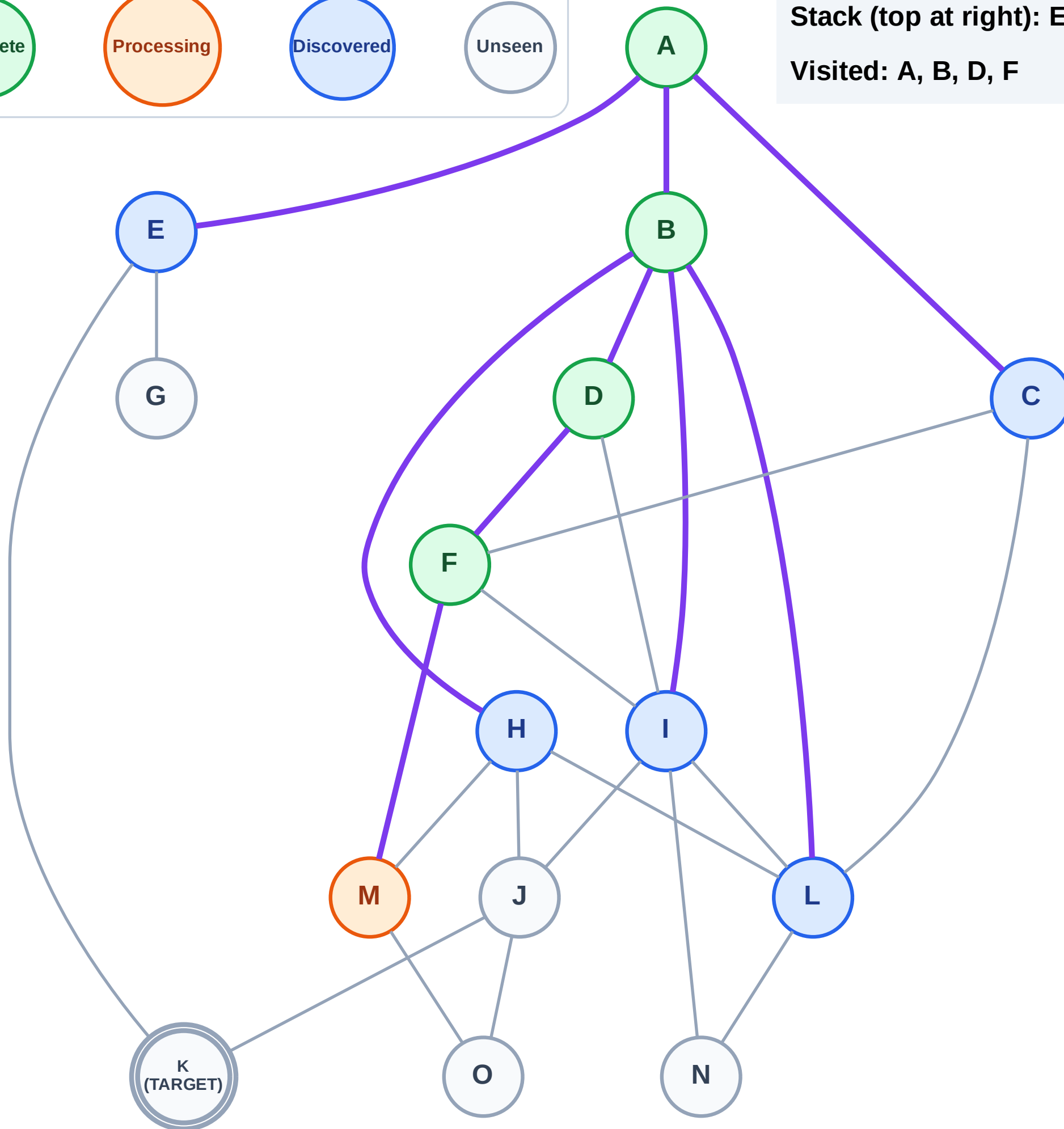
Step 10: Process node M

Legend



Stack (top at right): E | C | L | I | H

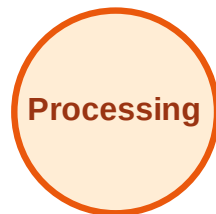
Visited: A, B, D, F



DFS Traversal

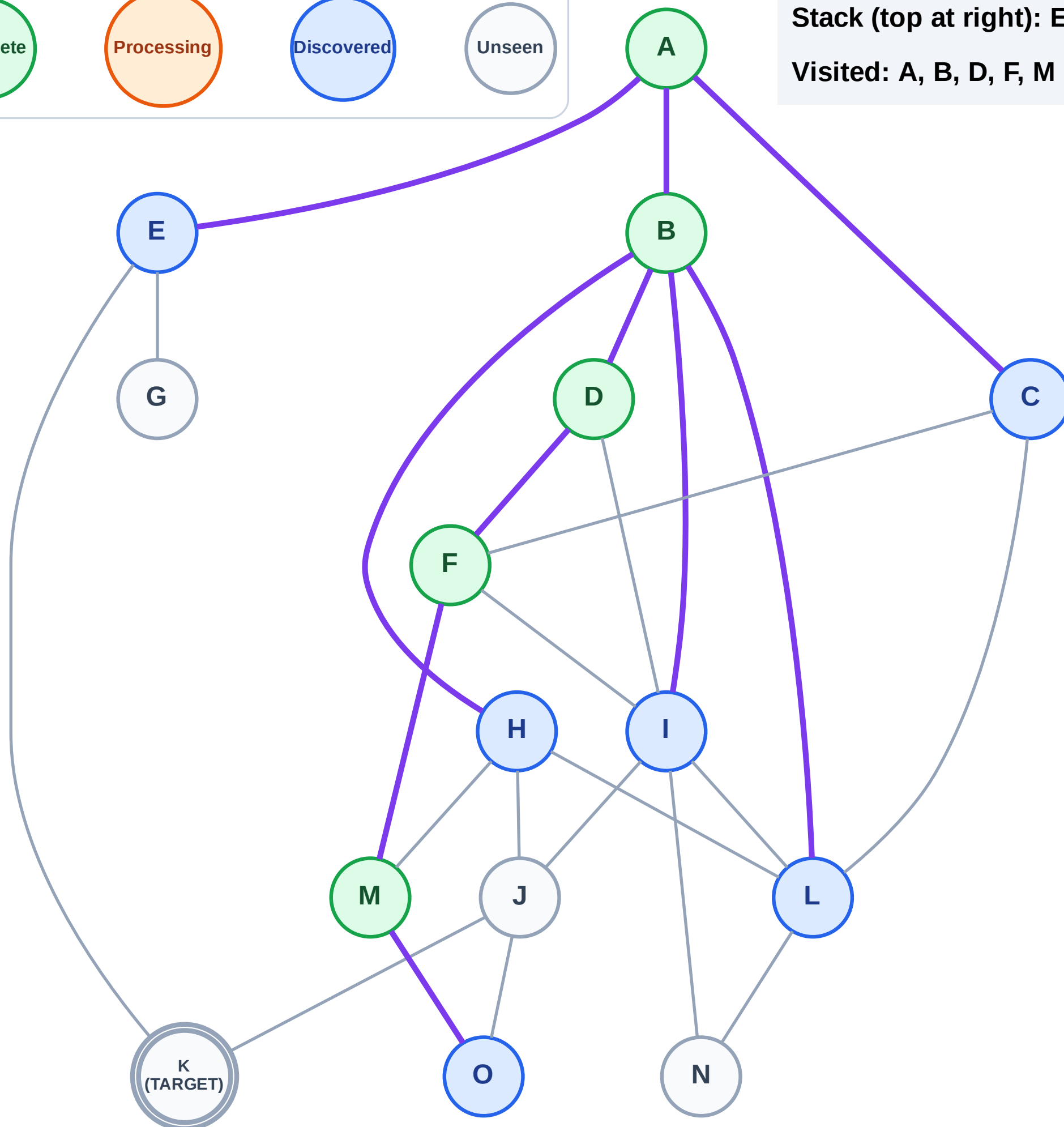
Step 11: Discover O from M

Legend



Stack (top at right): E | C | L | I | H | O

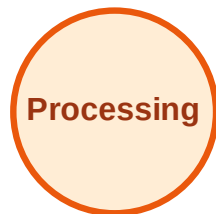
Visited: A, B, D, F, M



DFS Traversal

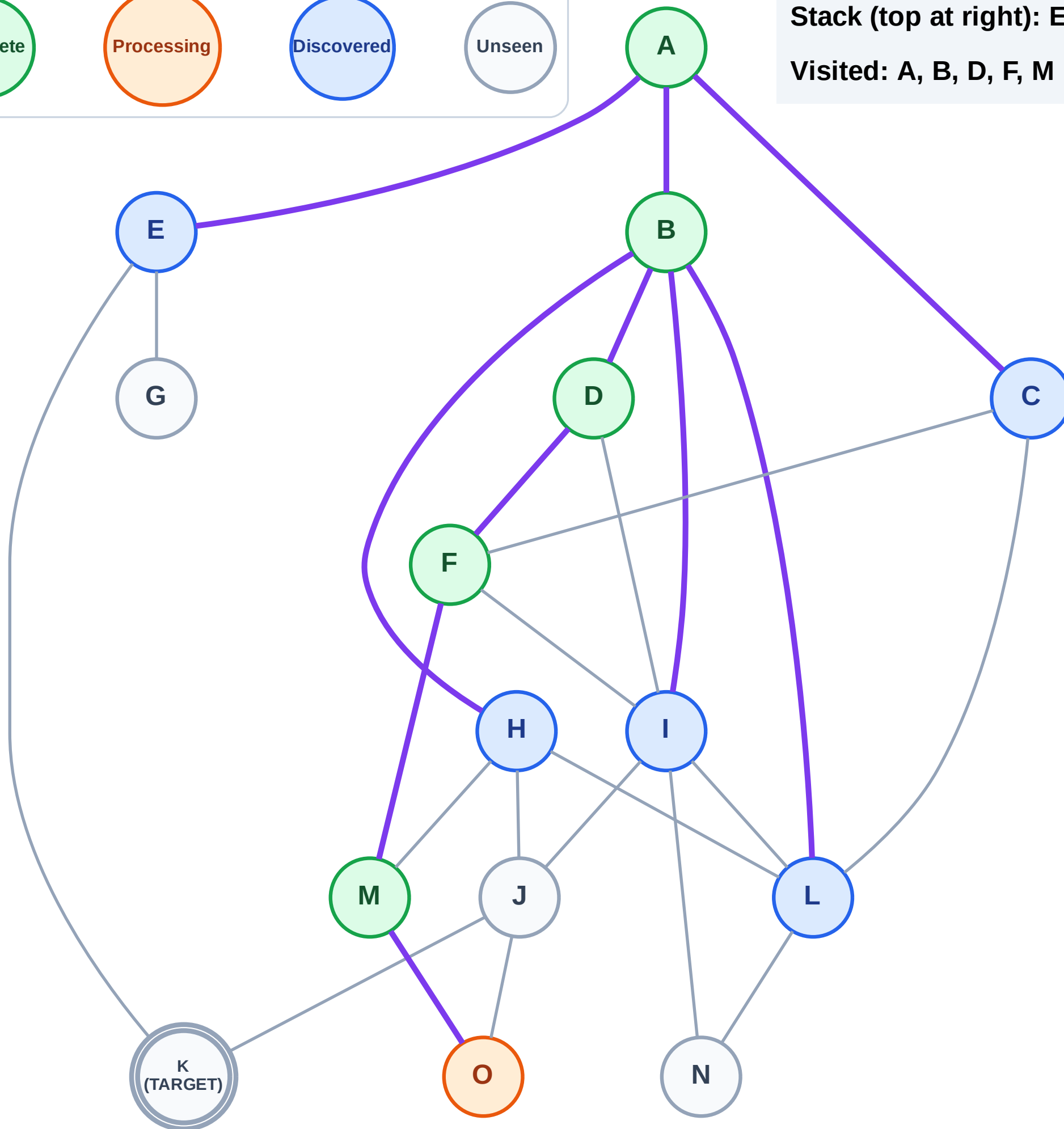
Step 12: Process node O

Legend



Stack (top at right): E | C | L | I | H

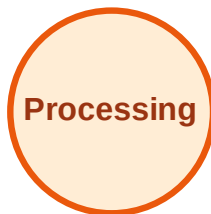
Visited: A, B, D, F, M



DFS Traversal

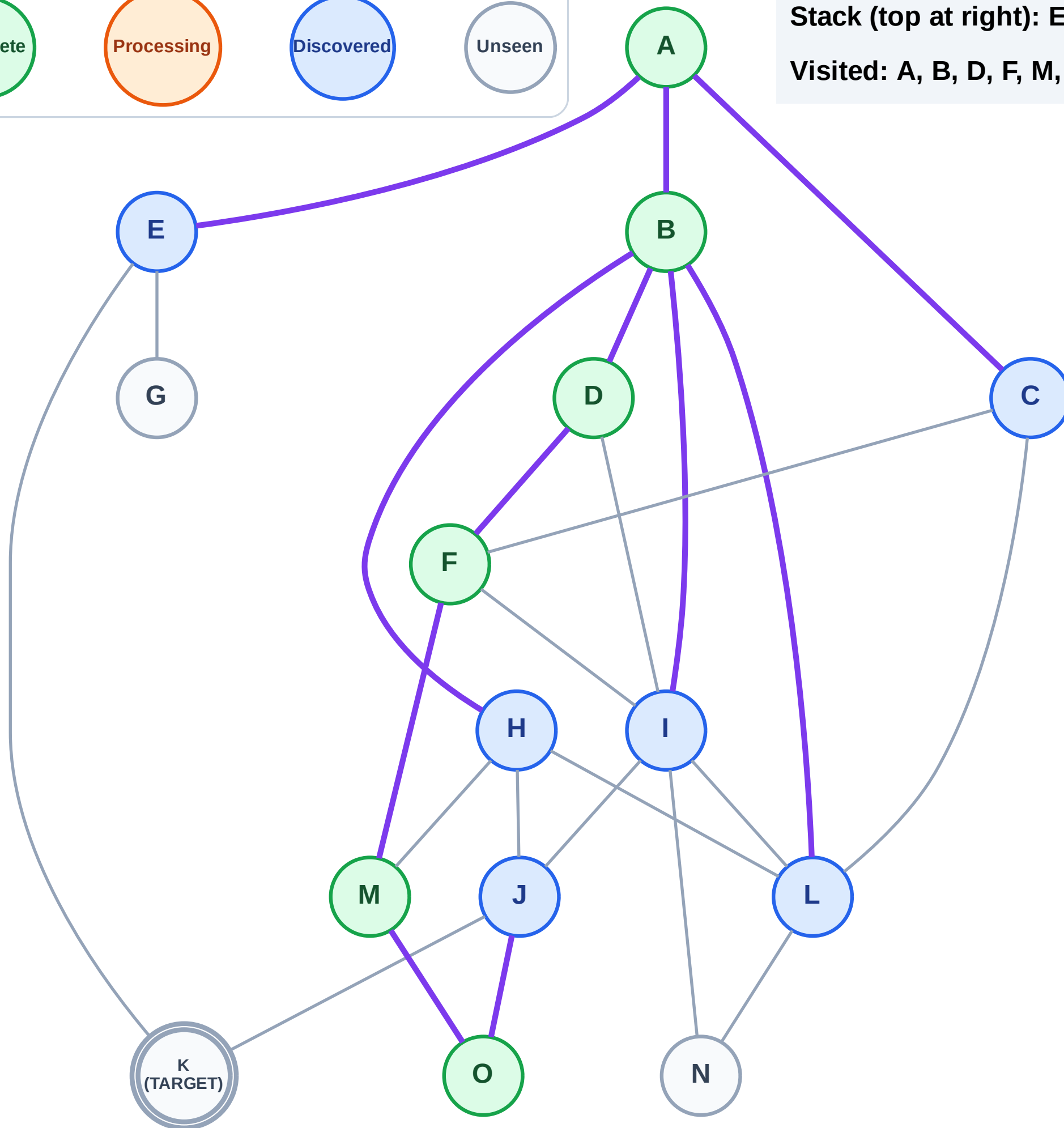
Step 13: Discover J from O

Legend



Stack (top at right): E | C | L | I | H | J

Visited: A, B, D, F, M, O



DFS Traversal

Step 14: Process node J

Legend

Complete

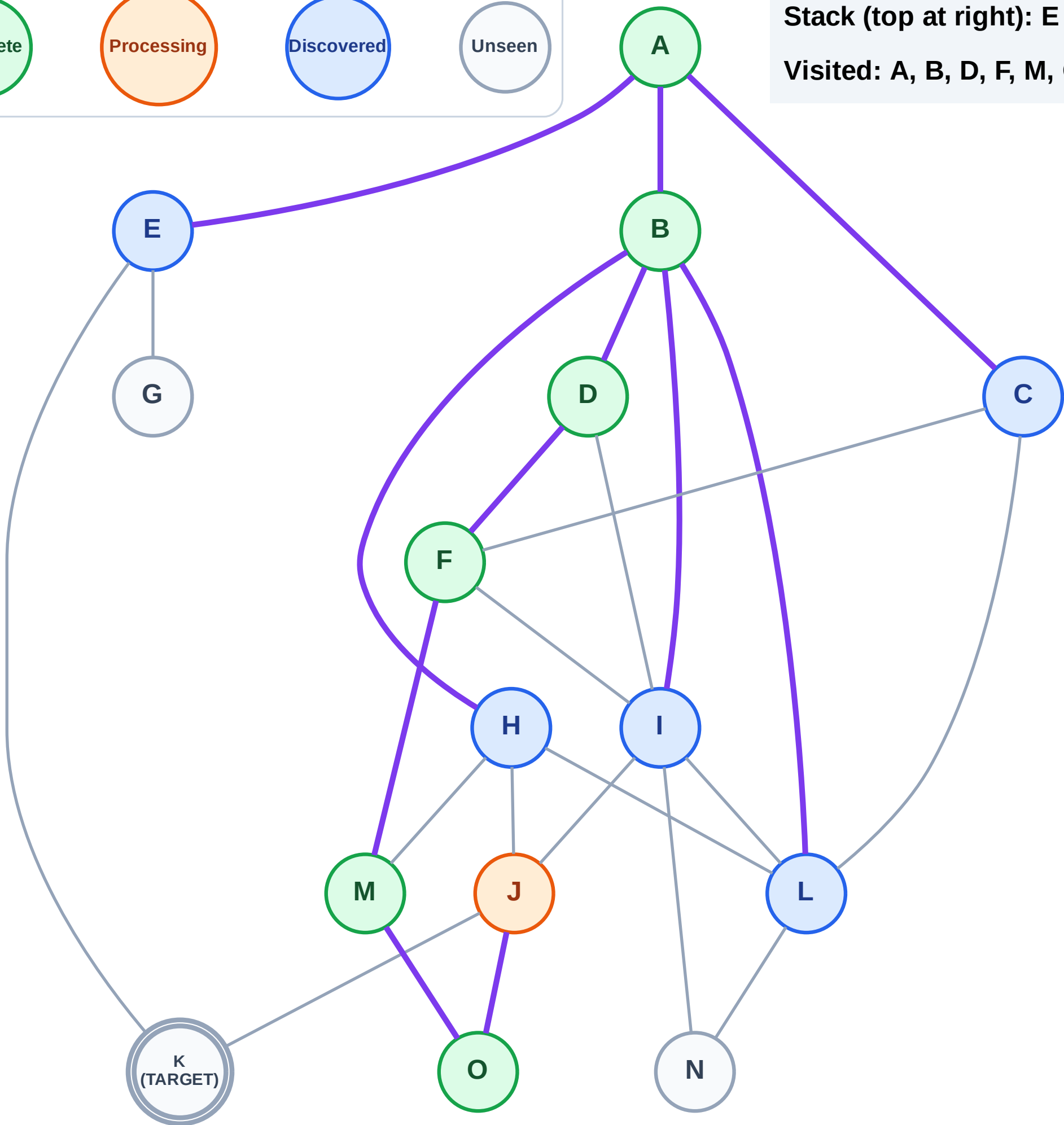
Processing

Discovered

Unseen

Stack (top at right): E | C | L | I | H

Visited: A, B, D, F, M, O



DFS Traversal

Step 15: Discover K from J

Legend

Complete

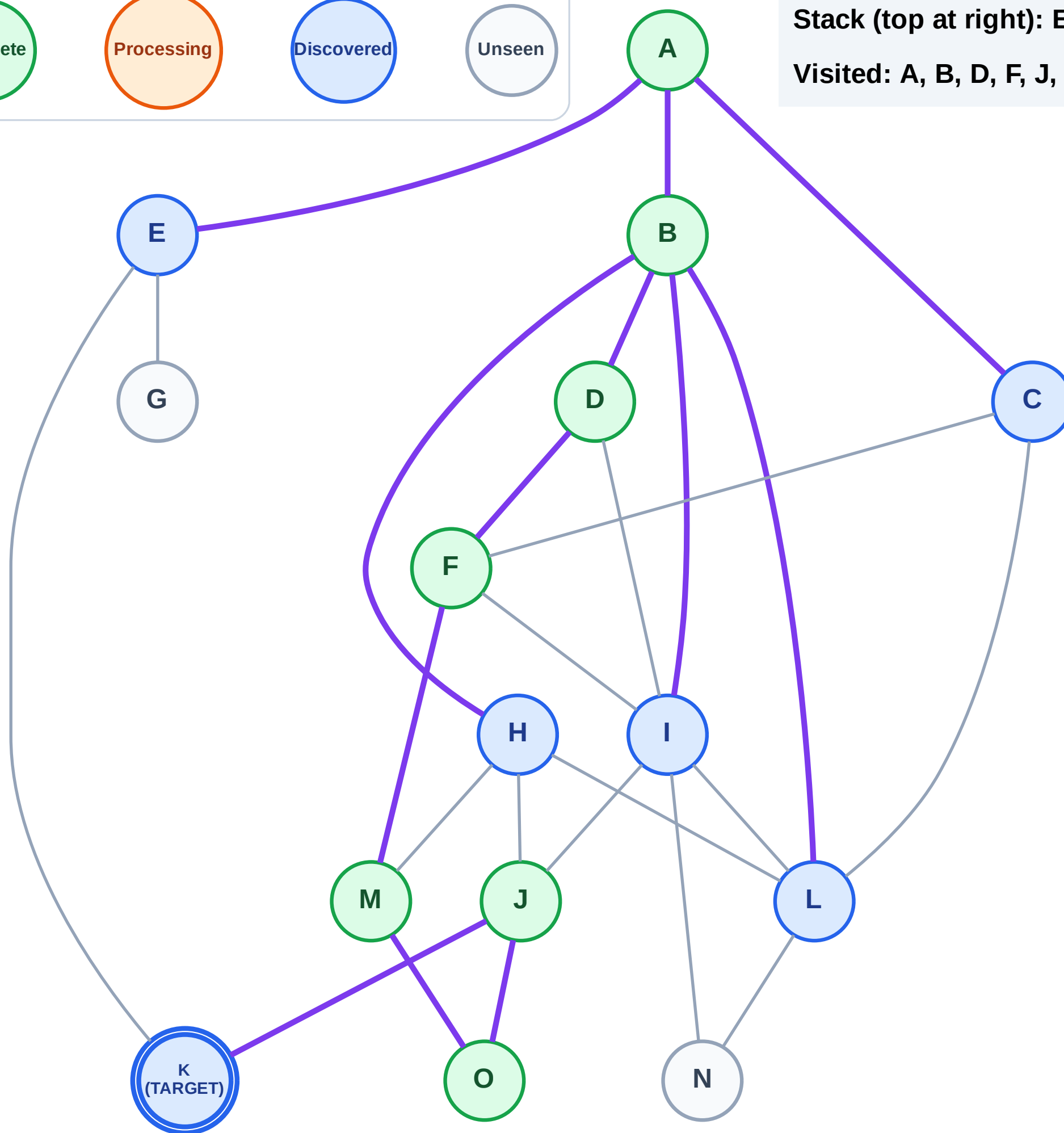
Processing

Discovered

Unseen

Stack (top at right): E | C | L | I | H | K

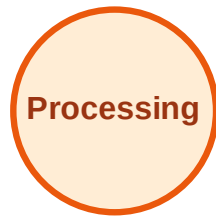
Visited: A, B, D, F, J, M, O



DFS Traversal

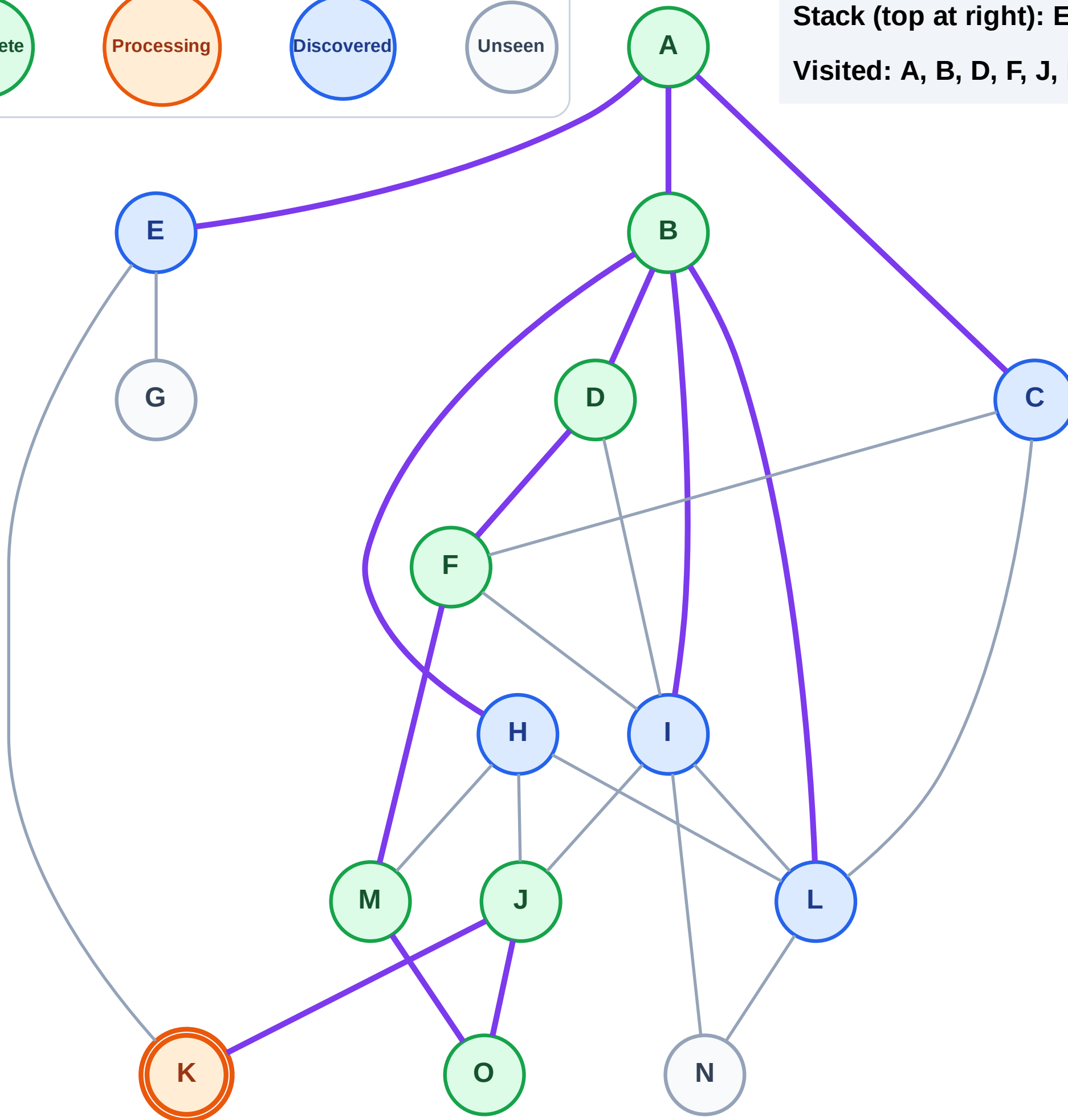
Step 16: Process node K

Legend



Stack (top at right): E | C | L | I | H

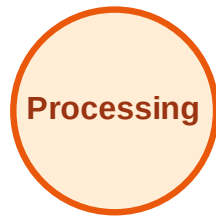
Visited: A, B, D, F, J, M, O



DFS Traversal

Step 17: TARGET NODE K REACHED! Path (7 hops): A → B → D → F → M → O → J → K

Legend



Stack (top at right): E | C | L | I | H

Visited: A, B, D, F, J, K, M, O

